Commitment and the Dynamics of Household Labour Supply*

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June 10, 2024

Abstract

The extent to which individuals commit to their partner for life has important implications. This paper develops a lifecycle collective model of the household, through which it characterises behaviour in three prominent alternative types of commitment: full, limited, and no commitment. We propose a test that distinguishes between all three types based on how contemporaneous and historical news affect household behaviour. Our test permits heterogeneity in the degree of commitment across households. Using recent data from the Panel Study of Income Dynamics, we reject full and no commitment, while we find strong evidence for limited commitment.

Keywords: Household behaviour; Intertemporal choice; Commitment; Collective model; Family labour supply; Dynamics; Wages; PSID.

JEL classification: D12; D13; D15; J22; J31.

*This paper develops and supersedes our previous work in Chiappori *et al.* (2020). We thank the editor, Francesco Lippi, three anonymous referees, and seminar and conference participants in the Structural Econometrics Group at Tilburg, Essex, KU Leuven, the ISR/PSID, Porto, Wuppertal, Barcelona Summer Forum (Income Dynamics and the Family & Structural Microeconometrics workshops), SOLE, SED, the Labour & Marriage Markets workshop at Aarhus, the Gender & Family Webinar by CY Cergy, and the Econometric Society North American Summer Meeting at UCLA for helpful comments. We benefited from many discussions with Jaap Abbring, Richard Blundell, Jeff Campbell, Sam Cosaert, George-Levi Gayle, Krishna Pendakur, Luigi Pistaferri, and Michèle Tertilt.

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1 Introduction

Large amounts of commitment are necessary for investing in common assets, for producing goods at home efficiently, and for pooling risk across family members. By contrast, limits to commitment typically induce investments in private assets, prevent the partners from economically abusing each other, and offer a way out from a bad marriage. In this paper, we develop a lifecycle collective model of the household, through which we characterise behaviour in three alternative regimes: full, limited, and no commitment. We show that current and past news affect behaviour differently in each case and we propose a test that distinguishes between all three. Using recent PSID data, we reject full and no commitment, while we find strong evidence for limited commitment with large heterogeneity across households.

Consider two or more parties who interact repeatedly sharing risk, such as spouses who offer each other intra-household insurance (Mazzocco, 2007; Lise and Yamada, 2019), village households who transfer goods or income among them (Townsend, 1994; Ligon *et al.*, 2002), workers who supply labour and firms that offer employment (Thomas and Worrall, 1988; Beaudry and DiNardo, 1991), or agents who trade assets (Kehoe and Levine, 1993; Alvarez and Jermann, 2000). The extent to which the parties commit to some future behaviour is clearly crucial as it shapes the degree of risk sharing between them.

This paper investigates this issue from a household perspective. We follow the by now dominant approach to household behaviour, the collective model (Chiappori, 1988, 1992), which relies on *cooperative* game theory. While in its static version this approach assumes that individuals reach Pareto-efficient agreements, its dynamic aspects are more complex. Although *ex post* efficiency is generally maintained, *ex ante* efficiency strongly depends on assumptions about the spouses' ability to commit. Following Mazzocco (2007), the literature has considered three alternative modes of commitment. At one extreme, full commitment is assumed: the spouses commit to a plan that disciplines the sharing of resources regardless of shocks that may affect them differently (e.g. Chiappori *et al.*, 2018). Full commitment has nice normative properties but its realism is disputable; in particular, there exists few (if any) legal ways to enforce this type of agreement. At the other extreme, no form of commitment is possible, and the relationship amounts to a sequence of short-term bargaining games (see Lundberg and Pollak, 2003, for a theoretical analysis, or Goussé *et al.*, 2017, for an empirical application). Clearly, such models result in a high level of dynamic inefficiency; in particular, they imply very strong restrictions on the ability to share risk within the group.

In the middle are limited commitment models, in which the spouses commit to a plan up to the point that some shock reduces one's individual welfare below their outside option (Mazzocco, 2007). They then renegotiate the plan or unilaterally switch to their outside

option as in the case of unilateral divorce (e.g. Voena, 2015). Such models have mostly been developed in a non-cooperative context (Thomas and Worrall, 1988; Kocherlakota, 1996); as such, they typically require an infinite horizon for the existence of non-trivial subgame perfect equilibria. Yet, they admit a straightforward, *cooperative* interpretation; namely, they are ex ante efficient in a second best sense, i.e. taking into account the limitations that exist from the spouse's ability to commit. Nevertheless, limited commitment models still assume quite a lot of commitment. For instance, if an agent's outside option increases (although not enough to become binding), she will not try to renegotiate a better outcome, even though the static bargaining solution will have evolved in her favour.[1]

Most studies of the dynamics of household behaviour make specific assumptions about commitment; their predictions are thus dependent on such assumptions. Yet, a few papers develop comparisons between models (Mazzocco, 2007; Lise and Yamada, 2019). Relying on a parametric framework, these works try to identify the Pareto weight in each period and contrast its dynamics between *full* and *non*-full commitment. The goal of our paper is to propose a more direct, reduced form approach by focusing on which variables affect behaviour in *each* (full, limited, no) commitment mode. Specifically, we do three things.

First, we characterise household behaviour in each commitment mode. We establish that shocks to the economic environment of the family affect behaviour differently in each case. The differences manifest via the Pareto weight on each person's preferences that disciplines the sharing of marital surplus. In full commitment, shocks, whether current or past, do not affect the Pareto weight, which remains constant over time or across states of the world. In limited commitment, current shocks may shift the Pareto weight if they trigger a renegotiation; this depends on the history of the couple, therefore also on past shocks, because individual welfare from marriage is a function of the past sharing of resources in the couple. In no commitment, by contrast, current shocks shift the Pareto weight continuously regardless of past shocks or circumstances. These restrictions on the Pareto weight translate into analogous restrictions on family labour supply, as we establish subsequently.

Second, we show that the set of variables that enter the full commitment Pareto weight (for which neither current nor historical information matters) is nested within the set that enters the no commitment weight (for which current information matters but historical information does not), which in turn is nested within the set that enters the limited commitment weight (for which all information matters). Thus the direction of nesting is not what one would expect by the names of the commitment regimes alone. Nesting, a common recursive

[1] In the non-cooperative setting of Thomas and Worrall (1988) or Kocherlakota (1996), partial commitment is enforced via the infinite horizon and the possibility of future retaliations in *any* period. In our finite horizon setting, only trivial subgame perfect equilibria exist, and partial commitment has to be assumed, in line with the mostly axiomatic approach of cooperative game theory. This point becomes clearer below.

form across modes (which we derive), and natural exclusion restrictions from current and past news allows us to devise a test that, for the first time, separates the three regimes.

Our test is about the presence of effects from current and past shocks, as well as the sign of such effects. In limited commitment, current shocks to distribution factors (variables that enter the Pareto weight) affect behaviour in a way that is tightly determined by the assignability of the shock. Consider a cash transfer to women (e.g. Armand *et al.*, 2020). If the transfer triggers a renegotiation, the female Pareto weight should increase, raising her leisure and reducing her labour supply. By contrast, the male weight would decrease, reducing his leisure and increasing his labour supply. These *asymmetric* effects reflect a power shift in the couple, where favourable news empower its recipient and simultaneously weaken their partner. The renegotiation, however, depends on the individual welfare from marriage today, which depends positively on the Pareto weight *until* today, which is itself determined by previous renegotiations. Therefore, the Pareto weight has memory and *past* shocks to distribution factors matter for current behaviour in the same asymmetric way between spouses as current shocks do. By contrast, history does not matter in no commitment and past shocks are bygones. Neither current nor past shocks matter in full commitment.

Third, we confirm that wages enter the Pareto weight naturally even though they are not conventional distribution factors (they also affect the budget set). We show that their bargaining effects (effects *through* the Pareto weight) are distinguishable from conventional income and substitution effects and can thus be used to test for commitment. This is appealing because wages are more easily available in household data than conventional distribution factors, they are assignable, and they typically vary considerably over time.

We implement our test empirically in a sample of married couples from the Panel Study of Income Dynamics in the US over the last two decades. Our main outcome of interest is individual labour supply in the household while our main source of shocks is male and female wage shocks. Our primary empirical exercise can thus be seen as one that investigates the dynamic effects of (current and past) wages on family labour supply.

We consistently reject full and no commitment. By contrast, we find strong evidence for limited commitment. Favourable shocks reduce one's own labour supply *and* increase the partner's; this is *simultaneously* true for current *and* historical shocks from multiple periods, precisely as limited commitment postulates. These effects, which cannot be explained on the basis of substitution, income, wealth, or tax adjustments, are consistent with power shifts in which favourable shocks improve the bargaining power of the recipient spouse. History matters under limited commitment, so shocks that shifted past bargaining power have lasting effects on behaviour in a very specific way. This is exactly what we find in the data.

The simplest form of our test can be implemented fairly easily in *reduced form*, with-

out parameterizing or estimating individual preferences. This is appealing because, on one hand, the test does not rely on a specific functional form for utility and, on the other hand, it can help quickly inform about commitment without simulating a rather involved dynamic model. However, estimation of parts of the *underlying structure* permits and reveals large heterogeneity in the degree of commitment across households. The overall evidence for limited commitment masks in fact that many couples exhibit full commitment (null bargaining effects), while others strongly exhibit limited commitment.

This paper contributes to the literature on household behaviour and, in particular, to its intertemporal aspects. Bargaining and in particular collective models have become the norm in this literature. Voena (2015) develops a limited commitment labour supply model to study the impact of unilateral divorce. Chiappori *et al.* (2018) develop a full commitment labour supply model to study the labour and marriage market implications of education choice. Lise and Yamada (2019) use a time use model with no commitment to study resource sharing. Foerster (2020) builds a household model with limited commitment to study how alimony affects parents' welfare.[2] While these excellent works select a priori the commitment technology available to agents (so their conclusions are conditional on that choice), we take a step back and *test* for the extent of commitment in married couples. As such, the closest paper to ours is the seminal work of Mazzocco (2007), who tests for full vs. non-full commitment based on whether current news affects consumption sharing. Lise and Yamada (2019) do similarly in a model with home production. They all find evidence for this and reject full commitment. While this is often seen as evidence for limited commitment, in reality these tests cannot formally separate no from limited commitment.[3] By contrast, the test we propose distinguishes between all three alternatives based on the additional role of historical information. Moreover, our test is not only about the presence of effects from current and past news but also about the sign of such effects that is strictly disciplined by theory.[4]

[2]Bronson (2014) and Mazzocco *et al.* (2014) use limited commitment (LC) lifecycle models to study education choices and household specialisation; Goussé *et al.* (2017) study home production under no commitment; Low *et al.* (2018) study welfare reforms with LC; Blasutto and Kozlov (2021) and Reynoso (2022) study how unilateral divorce affects cohabitation and the marriage market under LC; De Rock *et al.* (2023) study demand for housing in a LC setting. Fernández and Wong (2017) study marriage and divorce in a model in which spouses commit fully to resource sharing while married but cannot commit not to divorce. Chiappori and Mazzocco (2017) review the literature.

[3]Lise and Yamada (2019)'s model is, strictly speaking, one of no commitment. Empirically, they find that resource shares respond to large but not small shocks, which they interpret as indirect evidence for limited commitment because smaller shocks should in principle not shift the spouses' participation constraints. However, there are reasons unrelated to limited commitment why resource shares may not respond to small shocks, such as renegotiation costs under no commitment (appendix G). By contrast, the role of history formally distinguishes no and limited commitment, which is precisely the idea we put forward in this paper.

[4]Other tests include Townsend (1994) in the context of risk sharing in village economies, Blau and

The test is motivated by our characterisation of behaviour across commitment modes. The paper thus also relates to the macro and development literatures that study transfers without commitment (e.g. Coate and Ravallion, 1993; Kocherlakota, 1996; Ligon *et al.*, 2002; Dubois *et al.*, 2008).[5] Mazzocco (2007) and Adams *et al.* (2014) do similarly in a full information, collective household context without, however, considering all three regimes we analyze here.[6] However, there are alternative modes of behaviour, e.g. cooperative models with asymmetric information (Ashraf *et al.*, 2022), moral hazard (Kinnan, 2022), or general infinite horizon non-cooperative models, to which our test is not directly applicable.

The paper is finally related to the literature in labour economics that concerns the labour supply response to wages, particularly to the partner's wages, as in Lundberg (1985) and Hyslop (2001); Bellou and Kaymak (2012), Blundell *et al.* (2016) and Wu and Krueger (2021) are recent contributions. Our distinctive feature is the focus not only on responses to current wages but also on dynamic responses to spousal wages multiple periods in the past.

The paper develops as follows. Section 2 illustrates the main ideas in a simple setting. Section 3 presents the full lifecycle model and the complete characterisation of behaviour. Section 4 presents the formal test for commitment, section 5 discusses its implementation, and section 6 shows the results and several extensions. Section 7 concludes. An online appendix presents additional derivations and results.

2 Illustration of main ideas

This section illustrates the main ideas in a simple parametric framework with two periods $t \in \{t_1, t_2\}$, no discounting, and no savings. The next section relaxes these restrictions.

A household consists of two individuals, a male and a female, respectively subscripted by $j \in \{1, 2\}$. Each partner has logarithmic indirect utility $v_j(x_{jt}) = \ln x_{jt}$, where x_{jt} may be any assignable expenditure on goods or leisure; we will be more specific about this in section 3. Individual expenditure must satisfy the budget constraint $x_{1t} + x_{2t} = z_{1t} + z_{2t} = y_t$ in each period, where z_{jt} denotes the realisation of j's income at t. Each spouse has in each period an outside option $\bar{U}_{jt} = \phi(z_{jt})$, which may depend on his/her current income realisation. We will now derive individual consumption in each commitment case.

Under *full commitment*, the couple decides its full (contingent) allocation plan ex ante. Specifically, it solves a program of the form $\max_x v_1(x_{1t_1}) + v_1(x_{1t_2})$, subject to the budget

Goodstein (2016) in a household context, and Walther (2018) in a setting with production inefficiencies. Lafortune and Low (2020) show empirically that home ownership affects the degree of commitment.

[5] Our paper is also related to an expanding family literature in macroeconomics; see Doepke and Tertilt (2016) and Greenwood *et al.* (2017) for excellent reviews.

[6] Chiappori and Ekeland (2006) and Cherchye *et al.* (2007) characterise static collective behaviour.

constraint in each period t and an ex ante participation constraint for individual 2, given by $v_2(x_{2t_1}) + v_2(x_{2t_2}) \geq \bar{U}_2$. Here $\bar{U}_2$ is defined ex ante and may reflect the situation on the marriage market (as in Chiappori *et al.*, 2018) or stem from some initial bargaining, e.g., over the partners' expectations of future income.

If μ_2 denotes the Lagrange multiplier on the last constraint, the household program can be written as $\max_x \mu_1\big(v_1(x_{1t_1}) + v_1(x_{1t_2})\big) + \mu_2\big(v_2(x_{2t_1}) + v_2(x_{2t_2})\big)$, subject to the budget constraint in each period. μ_1 is the multiplier on spouse 1's utility. Equivalently, the household solves, in each period t, the program $\max_x \mu_1 v_1(x_{1t}) + \mu_2 v_2(x_{2t})$ subject to the budget constraint at t, with solutions given by

$$\frac{x_{jt_1}}{y_{t_1}} \text{ (period 1)} = \frac{x_{jt_2}}{y_{t_2}} \text{ (period 2)} = \frac{\mu_j}{\mu_1 + \mu_2}, \quad \text{for } j \in \{1, 2\}. \tag{1}$$

The crucial point is that the Pareto weights, μ_1 and μ_2, remain constant; in particular, they do *not* depend on the realisation of individual income in any period.[7] Under log utility, this implies a *constant* share of total income accruing to each individual.

Limited commitment requires deriving the *ex ante*, *second best* optimal contract. Different cases must be distinguished, depending on whether some participation constraint in marriage is violated (and, if so, in which period) by the full commitment solution. For instance, if the latter satisfies all participation constraints, then it coincides with the limited commitment solution. Alternatively, if income realisations result in a second period constraint being violated by the first best solution (1), then the second period Pareto weight will be modified such that the constraint becomes exactly binding; as a result, the second period allocation will be affected by the second period income shocks.

A complete resolution is left to section 3; here, we simply focus on a particularly interesting case when a first period participation constraint is violated. For brevity, we let the outside options at t_2 be very low (so that second period constraints can be omitted). The ex ante, second best program can be written as $\max_x \mu_1\big(v_1(x_{1t_1}) + v_1(x_{1t_2})\big) + \mu_2\big(v_2(x_{2t_1}) + v_2(x_{2t_2})\big)$, subject to the budget constraint in each period and the first period participation constraints $v_j(x_{jt_1}) + v_j(x_{jt_2}) \geq \phi(z_{jt_1})$, for $j \in \{1, 2\}$. Suppose, now, that the full commitment solution (1) violates one participation constraint – say, of $j = 1$, namely $\ln \frac{\mu_1}{\mu_1+\mu_2} y_{t_1} + \ln \frac{\mu_1}{\mu_1+\mu_2} y_{t_2} < \phi(z_{1t_1})$, while the constraint of $j = 2$ is satisfied. At the *second best* solution, spouse 1's constraint must be binding; its Lagrange multiplier $\nu(z_{1t_1})$ will be positive (while that on spouse 2's constraint is nil) and depend on the income realisation z_{1t_1}. The household program then becomes $\max_x (\mu_1 + \nu(z_{1t_1}))\big(v_1(x_{1t_1}) + v_1(x_{1t_2})\big) + \mu_2\big(v_2(x_{2t_1}) + v_2(x_{2t_2})\big) - \nu(z_{1t_1})\phi(z_{1t_1})$ subject

[7] Individual expenditure thus depends only on total income, not its individual realisations. This property, called the *mutuality principle*, characterises efficient risk sharing, enabled by the full commitment assumption.

to the budget constraint in each period (details in appendix A). Its solutions are

$$\frac{x_{1t_1}}{y_{t_1}}\ (\text{period 1}) = \frac{x_{1t_2}}{y_{t_2}}\ (\text{period 2}) = \frac{\mu_1 + \nu(z_{1t_1})}{\mu_1 + \nu(z_{1t_1}) + \mu_2},$$
$$\frac{x_{2t_1}}{y_{t_1}}\ (\text{period 1}) = \frac{x_{2t_2}}{y_{t_2}}\ (\text{period 2}) = \frac{\mu_2}{\mu_1 + \nu(z_{1t_1}) + \mu_2},$$

and the expenditure shares at t_2 depend on the realisation of *period 1* income z_{1t_1} through $\nu(z_{1t_1})$. Technically, spouse 1's weight increases from μ_1 to $\mu_1 + \nu(z_{1t_1})$ in period 1 following the binding constraint, and this change persists in the absence of new violations. Note that the model assumes quite a lot of commitment; the period 2 outside option may have increased, but not enough to bind, so the spouses do not try to further renegotiate. It is precisely this level of commitment that generates the memory property we describe.

Under *no commitment*, in the definition of 'bargaining in marriage' of Lundberg and Pollak (2003) and Pollak (2019), the spouses bargain in each period. Cooperative bargaining theory requires that, in each period, the allocation be Pareto efficient in the *ex post* sense; therefore the household in period t maximises a weighted sum of utilities $\mu_{1t}v_1 + \mu_{2t}v_2$, giving the solution

$$\frac{x_{jt}}{y_t} = \frac{\mu_{jt}}{\mu_{1t} + \mu_{2t}}, \quad \text{for } j \in \{1, 2\} \text{ and } t \in \{t_1, t_2\}.$$

In order to determine the Pareto weights μ_{jt} and the allocation of resources in each period, the model is solved backwards. In period 2, the allocation, as summarised by the weights μ_{jt_2}, depends only on the period 2 bargaining framework, i.e. total income y_{t_2} and the second period reservation utilities $\phi(z_{1t_2})$ and $\phi(z_{2t_2})$.[8] In period 1, μ_{jt_1} and the allocation depend on y_{t_1}, $\phi(z_{1t_1})$ and $\phi(z_{2t_1})$, as well as on the expectations of individual future incomes $\tilde{z}_{1t_2}$, $\tilde{z}_{2t_2}$ given z_{1t_1}, z_{2t_1}, but, obviously, not on the *realisations* of z_{1t_2} and z_{2t_2}.

Two remarks are due here. First, from a non-cooperative viewpoint, in any *finite horizon* model of this type, the *only* subgame perfect equilibrium leads to the above no commitment outcome. Second, the period 2 allocation does *not* vary with period 1 incomes.[9] In period 2 the spouses play a static, cooperative bargaining game, the outcome of which depends only on *current* total income y_{t_2} and outside options, as summarised by $\mu_{jt_2}\big(y_{t_2}, \phi(z_{1t_2}), \phi(z_{2t_2})\big)$. None of the past period's incomes matter anymore – bygones are bygones.

In summary, under full commitment, idiosyncratic income realisations cannot affect expenditure shares in any period. Under limited commitment, expenditure shares may reflect

[8] The exact form of $\mu_{jt_2}\big(y_{t_2}, \phi(z_{1t_2}), \phi(z_{2t_2})\big)$ depends on the specific bargaining concept at play, on which no assumptions are made beyond cooperation.

[9] Technically, this relies on the existence of a unique Nash equilibrium in the stage game. This would be violated if, e.g., one's utility v_j depends on the partner's expenditure x_{-jt}, a strong form of 'caring'. We assume this type of extreme complementarity away; we thank an anonymous referee for pointing this out.

current realisations (if a current participation constraint is violated) and past ones (if those have modified the past Pareto weight but current realisations have left it intact). Under no commitment, idiosyncratic realisations in each period influence the expenditure shares for that period, but past ones influence neither current nor future expenditure shares.

3 Household lifecycle behaviour

We now switch to our full setting, namely the lifecycle collective model in which forward-looking spouses make consumption/hours choices subject to idiosyncratic wage risk.

A household consists of a male ($j = 1$) and a female ($j = 2$) spouse, who get married at time $t = 0$ and live for $\bar{t}$ periods. In each period $t \in \{0, \ldots, \bar{t}\}$, each person enjoys utility from joint consumption q_t and disutility from labour hours h_{jt}, as per individual preferences $u_j(q_t, h_{jt})$. We assume that u_j has continuous first/second partial derivatives with $u_{j[q]} > 0$, $u_{j[h]} < 0$ (disutility of work), and $u_{j[qq]} < 0$, $u_{j[hh]} < 0$ (concavity); utility is *not* quasi-linear, a class of preferences for which we find no empirical support anyway.

The couple's budget constraint, common across commitment alternatives, is given by

$$(1 + r)a_t + \tau(y_t) = q_t + a_{t+1}, \quad \forall t, \tag{2}$$

where a_t is common financial assets at the start of the period and r is the deterministic interest rate.[10] τ maps gross household earnings $y_t = w_{1t}h_{1t} + w_{2t}h_{2t}$ into disposable income y_t^D, accounting for joint taxation and benefits (e.g. EITC). w_{jt} is the (individual, stochastic) price of an hour of market labour and $W_t = \{w_{1t}, w_{2t}\}$ is the set of wages in the couple. We assume the spouses hold identical beliefs about future stochastic elements.

Both τ and individual utility u_j depend on demographics, such as fertility. We suppress those here for brevity (we thus fix a household type), but we do account for them in the empirical application. Appendix A shows the role of demographics in the model explicitly.

3.1 Commitment modes

The next three subsections present the collective model in the three alternative commitment regimes. For now, we disregard divorce and thus model behaviour conditional on continued marriage. Divorce is discussed in appendix A.

[10] The extension to risky assets is straightforward and does not affect the subsequent discussion. Assets in marriage are jointly held reflecting the community property regime in the US (Mazzocco, 2007).

3.1.1 Full commitment

Upon marriage, the individuals commit fully to all future but state-contingent allocations of resources between them. In other words, they commit at $t = 0$ to a plan that disciplines their actions in the future. Choices made under full commitment are therefore ex ante efficient and can be represented by the solution to the following problem at $t = 0$:

$$V_0^{\text{FC}}(\Omega_0) = \max_{\{C_t\}_{0\leq t\leq \bar{t}}} \mu_1(\Theta_0)\left(\mathbb{E}_0 \sum_{t=0}^{\bar{t}} \beta^t u_1(q_t, h_{1t})\right) + \mu_2(\Theta_0)\left(\mathbb{E}_0 \sum_{t=0}^{\bar{t}} \beta^t u_2(q_t, h_{2t})\right) \tag{3}$$

subject to the budget constraint (2) $\forall t$,

where $C_t = \{q_t, h_{1t}, h_{2t}, a_{t+1}\}$ is the set of household choice variables in period t. For brevity, we do not show their dependence on contingent states – we relegate this to appendix A.[11]

The Pareto weights μ_1 and μ_2 are the utility weights the household places on each person's preferences at marriage. They determine ex ante the relative allocation of resources between spouses. Θ_0 is the set of variables that affect the Pareto weights; because the weights are determined at marriage, it follows that Θ_0 must only include information known or predicted at the time the household is formed, that is, at time $t = 0$.[12]

The expectations, common between spouses as we assume throughout, are taken over the stochastic elements of future states in Ω_t, $t > 0$, such as future wages. Generally, the state space differs across commitment alternatives. We have purposefully not defined what explicitly goes into Ω but we will return to this point in the following sections.

3.1.2 Limited commitment

When an individual can unilaterally walk away from their partner (e.g. unilaterally divorce), not all future allocations are feasible in all states of the world. Certain plans may make one person better off outside the household. Assuming there is always a positive marital surplus to be shared, the household must make sure to not implement those plans.

Upon marriage, the spouses commit to future and state-contingent allocations of resources up to the point that one's marital participation constraint is violated. Choices under limited commitment are ex ante second-best efficient (ex ante efficient subject to participation

[11] We maintain a common discount factor between spouses; see Adams *et al.* (2014) for a generalisation.

[12] The discussion that follows only requires that some variables known at $t = 0$ affect the Pareto weights at marriage; it does not require to specify why exactly this happens. One may think that the weights arise from a bargaining game played at marriage or from equilibrium conditions in the marriage market (e.g. Chiappori *et al.*, 2018). In either case, information at $t = 0$ (e.g. variables summarizing the marriage market) affects the Pareto weights; we thus allow μ_1 and μ_2 to depend on Θ_0 to reflect, in reduced form, such a link.

constraints) and can be represented by the solution to the following problem at $t = 0$:

$$V_0^{\text{LC}}(\Omega_0) = \max_{\{C_t\}_{0\leq t\leq \bar{t}}} \mu_1(\Theta_0)\left(\mathbb{E}_0 \sum_{t=0}^{\bar{t}} \beta^t u_1(q_t, h_{1t})\right) + \mu_2(\Theta_0)\left(\mathbb{E}_0 \sum_{t=0}^{\bar{t}} \beta^t u_2(q_t, h_{2t})\right) \tag{4}$$

subject to the budget constraint (2) $\forall t$, and the participation constraints:

$$\nu_{jt} \;:\; \underbrace{\mathbb{E}_t \sum_{\tau=t}^{\bar{t}} \beta^{\tau-t} u_j(q_\tau, h_{j\tau})}_{\substack{\text{individual inside value} \\ \text{from marriage at } t}} \;\geq\; \underbrace{\widetilde{V}_{jt}(\Omega_{jt})}_{\substack{\text{individual outside value} \\ \text{at } t \text{ (e.g. single/divorced)}}} \qquad \forall t > 0, j \in \{1, 2\},$$

where $C_t = \{q_t, h_{1t}, h_{2t}, a_{t+1}\}$ is the set of household choice variables in period t and $\widetilde{V}_{jt}(\Omega_{jt})$ is the reservation utility of person j at t, defined over $\Omega_{jt} \subseteq \Omega_t$ – more on this below.

The participation constraints, one per individual and time period, ensure that each person enjoys at least as much value inside their joint household as they can possibly get from their outside option, i.e. by walking away from the relationship. In other words, the participation constraints ensure individual rationality in the relationship. The constraints consist of two parts, the *inside* (left hand side) and *outside* (right hand side) values, defined on the basis of the forward-looking continuation value of each option.

The *inside* value of person j reflects the share of marital surplus that accrues to him/her given the choices made by the household in the period. Naturally, this value varies with the applicable Pareto weight in the period: a larger Pareto weight on person j implies household choices more tailored to j's tastes, thus accruing a larger share of marital surplus to him/her, and vice versa. This link between the Pareto weight and the individual inside value from marriage disciplines the dynamics of the Pareto weight in limited commitment, a point to which we return in the next sections.

The *outside* value of person j reflects how he/she may fare in life outside the current relationship; its precise form depends on specific assumptions about that situation. For instance, we may, as in Voena (2015), assume that the outside option is divorce, and that the corresponding value is the present value of future expected utility of a single/divorced person – in which case we have:

$$\widetilde{V}_{jt}(\Omega_{jt}) = \max_{\{q_{j\tau}, h_{j\tau}, a_{j\tau+1}\}_{\tau=t,\ldots,\bar{t}}} \mathbb{E}_t \sum_{\tau=t}^{\bar{t}} \beta^{\tau-t} \widetilde{u}_j(q_{j\tau}, h_{j\tau})$$

$$\text{subject to } \; (1+r)a_{j\tau} + \tau(w_{j\tau}h_{j\tau}) = q_{j\tau} + a_{j\tau+1}, \; \forall\tau, \text{ and } a_{1t} + a_{2t} = a_t.$$

Here, we let preferences depend on marital status (i.e. $\tilde{u}_j$ may differ from u_j) to capture marital preference shifts. $\widetilde{u}_j$ may include stochastic elements that reflect, in reduced form,

the utility flow from possible future remarriage or grieving following the breakup.

Other interpretations are however possible and the precise form of $\widetilde{V}_{jt}$ does not matter for the subsequent discussion. The key aspect is that $\widetilde{V}_{jt}$ is defined over the (single's) state space $\Omega_{jt} \subseteq \Omega_t$. Three distinct sets of variables enter Ω_{jt}, whose role we describe below: the individual wage rate w_{jt} (wages matter for j's budget as single and for his/her labour market and possibly remarriage prospects), distribution factors Z_{jt}, and marital assets a_t. The expectations are taken over the stochastic elements of future states in $\Omega_{j\tau}$, $\tau > t$.

Distribution factors Z_{jt} are exogenous stochastic variables that affect the singles' lifecycle prospects but not preferences or the couple's budget set conditional on household income (Bourguignon *et al.*, 2009).[13] These variables thus affect the outside but not the inside options in the household. The difference between Z_{jt} and Θ_0 is that the former variables vary during the course of the relationship while the latter do not. As the variables in Z_{jt} vary stochastically over time, the single's outside value varies in response, which may make a participation constraint occasionally bind. To satisfy the constraint, the inside value of the constrained spouse must adjust, thus creating a link from the time-varying distribution factors to the choices made in the couple as we illustrate subsequently.

Upon household break-up, financial wealth accumulated during marriage is split between partners according to some fixed rule. Marital assets thus determine the wealth that a newly single possesses upon break-up, so wealth a_t enters j's outside value at t. This renders the outside options endogenous to choices made during marriage. A couple in our setting makes savings choices accounting for the implications of those choices for the outside options, in addition to the standard lifecycle/precautionary motives present also in full commitment.[14]

3.1.3 No commitment

While full and limited commitment feature some form of commitment at marriage to a future plan (the plan contingent on the Pareto weights at marriage and, in the case of limited commitment, the participation constraints), no commitment features no such marital 'contract'. Upon marriage, the spouses do not commit to a future plan, that is, they do not guarantee each other a certain or minimum allocation of resources.

Without commitment, new information that arises over time changes the division of marital surplus between spouses according to the bargaining game they play. Choices under

[13] Examples include relative non-labour incomes (e.g. Thomas, 1990; Attanasio and Lechene, 2014), the sex ratio in the marriage market (Chiappori *et al.*, 2002), or divorce and property division laws (Voena, 2015).

[14] The dependence of the outside options on wealth may move the household away from second-best efficiency; for example, the couple may overinvest in financial assets to improve their outside options. Chiappori and Mazzocco (2017) have a lengthy discussion of this as well as of an interesting opposite case.

no commitment can be represented by the solution to the following problem at $t = 0$:

$$V_0^{\text{NC}}(\Omega_0) = \tag{5}$$
$$\max_{\{C_t\}_{0\leq t\leq \bar{t}}} \left(\mathbb{E}_0 \sum_{t=0}^{\bar{t}} \beta^t \mu_1(\Theta_0, W_t, Z_t, a_t) u_1(q_t, h_{1t}) \right) + \left(\mathbb{E}_0 \sum_{t=0}^{\bar{t}} \beta^t \mu_2(\Theta_0, W_t, Z_t, a_t) u_2(q_t, h_{2t}) \right)$$
subject to the budget constraint (2) $\forall t$,

where $C_t = \{q_t, h_{1t}, h_{2t}, a_{t+1}\}$ is the set of household choice variables in period t.

The underlying premise of no commitment is that the spouses engage in some form of repeated bargaining over the marital surplus in a way that reflects the prevailing economic environment, captured by wages W_t, distribution factors $Z_t = \{Z_{1t}, Z_{2t}\}$, and wealth a_t. As new information arises over time, a person's bargaining position shifts given the bargaining game played by the spouses. The variables in Θ_0 are fixed after marriage but they enter the Pareto weights because they may influence the type of game the couple plays.

Choices under no commitment are ex ante inefficient. This is because ex ante efficiency, at least in the first-best sense, implies that there exist no time-contingent transfers that improve both spouses' expected utilities (Browning *et al.*, 2014). This requires the Pareto weights be the same over time, which is clearly not the case here. Choices are dynamically inefficient also because the bargaining weights depend on wealth. The spouse whose future weight increases more with wealth has an incentive to overinvest in assets, thus creating inefficiencies over time. Whether such inefficiency appeals to couples is ultimately an empirical question that our test for commitment helps address. Nevertheless, choices in a given period are ex post efficient in the sense that they maximise a weighted sum of individual period utilities.

Our representation of no commitment (in particular *how* new information impacts the Pareto weights) is arguably abstract and lacks the precise microfoundations of full or limited commitment. Nevertheless, this abstractness enables it to be consistent with several popular underlying structures, such as resource allocations with Nash bargaining over the marriage market (Goussé *et al.*, 2017), household sharing when labour market shocks shift the balance of power (Lise and Yamada, 2019), equilibrium allocations with bargaining over a default arrangement (Kato and Ríos Rull, 2023), and other.

3.2 Common recursive formulation

The household problem in each commitment mode is a dynamic planning problem over the allocation of resources between spouses and across periods of time. Following Marcet and

Marimon (2019), we can recast each problem in the common recursive form:

$$V_t(\Omega_t) = \max_{C_t} \ \mu_{1t}u_1(q_t, h_{1t}) + \mu_{2t}u_2(q_t, h_{2t}) + g_t(a_t) + \beta \mathbb{E}_t V_{t+1}(\Omega_{t+1}) \tag{6}$$

subject to the budget constraint (2), and

restrictions on the Pareto weights μ_{jt}, $j \in \{1, 2\}$, defined subsequently,

with details reported in appendix A. In each period, the household maximises a weighted sum of period utilities, an appropriate continuation value, and an additional term described below. Expectations are over the stochastic elements in Ω_{t+1}, given the realisation of Ω_t.

The additional term, $g_t(a_t)$, aggregates the singles' endogenous outside options in limited commitment. It is given by $g_t(a_t) = -\nu_{1t}\widetilde{V}_{1t}(w_{1t}, Z_{1t}, a_t) - \nu_{2t}\widetilde{V}_{2t}(w_{2t}, Z_{2t}, a_t)$ in limited commitment, where ν_{jt} is the Lagrange multiplier on spouse j's participation constraint at t, and by $g_t(a_t) = 0$ otherwise. $g_t(a_t)$ highlights that a couple makes savings choices in limited commitment taking into account the effect of those choices on the partners' outside options.

The solution to (6) is a set of time-consistent[15] optimal policy functions $q_t^*(\Omega_t)$, $h_{jt}^*(\Omega_t)$, $\forall j$, and $a_{t+1}^*(\Omega_t)$, which depend on the state space Ω_t in each commitment alternative. Our test for commitment relies on estimating equations derived directly from (6) and its corresponding policy functions, given restrictions that the Pareto weight imposes on the state space in each case. The crucial point in (6) is that there is a pair of applicable Pareto weights μ_{1t} and μ_{2t} in each period, the dynamics of which we will now characterise.

3.3 Characterisation of the Pareto weight and the state space

Bargaining power is relative inside the household since the sum $\mu_{1t} + \mu_{2t}$ can be normalised to a constant. Therefore, we subsequently refer to the Pareto weight in singular. Moreover, any variable that affects one person's Pareto weight must simultaneously and mechanically enter and affect the partner's weight in the opposite direction.[16]

Full commitment. The Pareto weight is determined at marriage as a function of the initial bargaining variables Θ_0. The weight remains constant over time, namely

$$\mu_{jt} = \mu_j(\Theta_0), \qquad j \in \{1, 2\}, \forall t,$$

as we show in appendix A. The variables in Θ_0 vary in the cross-section reflecting the couple's characteristics at marriage, local marriage market conditions, or other heterogeneity that the

[15] The policy functions depend on time because the horizon is finite.

[16] One can show this formally by explicitly including the restriction $\mu_{1t} + \mu_{2t} = \textit{constant}$ in program (6).

individuals base their initial bargaining on. Therefore, $\mu_j(\Theta_0)$ varies in the cross-section of households. The bargaining variables in Θ_0 remain fixed for $t > 0$ after marriage, so $\mu_j(\Theta_0)$ also remains fixed within a given family over time. The initial weight at marriage thus serves as the spouses' intra-household bargaining power over their entire lifecycle.

The time and state invariance of the Pareto weight (conditional on Θ_0) is a well-known implication of first-best efficiency (Browning *et al.*, 2014). Intuitively, once the initial bargaining weight is set at marriage, future shocks (for example, shocks to wages) do not change the allocation of marital surplus between spouses, who fully share any idiosyncratic risk between them. Of course efficiency requires that the spouses exploit the economic opportunities that arise from variation in wages or other shocks; but with full commitment, those effects remain compatible with ex ante efficiency and the Pareto weight does not change in response. This implies that policies that seek to empower, say, women, e.g. cash transfers targeted to women, cannot affect the division of marital surplus if they are implemented after marriage. The only policies that matter under full commitment are those that affect Θ_0.

Limited commitment. The Pareto weight is given by:

$$\begin{aligned} & \mu_{jt} = \mu_{jt-1} + \nu_{jt}, && j \in \{1,2\}, \forall t > 0, \\ \text{with} \quad & \mu_{j0} = \mu_j(\Theta_0), && j \in \{1,2\}, \end{aligned}$$

where ν_{jt} is the Lagrange multiplier on spouse j's participation constraint at t. We derive this in appendix A. The Pareto weight shifts when the continuation of the previous allocation of resources, summarised by the past weight μ_{jt-1}, violates one's participation constraint. In such case, the constrained spouse's weight jumps by $\nu_{jt} > 0$, i.e. the multiplier on the binding constraint. If no participation constraint binds, then $\nu_{jt} = 0$, and the Pareto weight remains unchanged. Whether a participation constraint binds depends on the variables underlying the constraint, so we may write $\nu_{jt} \equiv \nu_j(W_t, Z_t, a_t, \mu_{jt-1})$ as we explain below.

Consider the distribution factors. As $Z_t = \{Z_{1t}, Z_{2t}\}$ shift the outside options, a person's participation constraint may bind for some realisation of Z_t. To relax the constraint, the constrained person's bargaining power increases by ν_{jt}, which shifts household decisions towards her preferences and improves her inside value. The increase in power is the smallest possible that makes j indifferent between staying in the relationship and leaving. This follows from second-best efficiency as analyzed in Ligon *et al.* (2002).[17]

Wages also impact the participation constraints, though their workings are more nuanced.

[17]The present analysis is conditional on the spouses remaining married (marital surplus remains positive), so at most one participation constraint can bind in a given period (Kocherlakota, 1996). Therefore, following the increase in j's power, there exists a feasible allocation at which her partner's constraint is also satisfied.

Unlike distribution factors in Z_t, wages simultaneously affect j's outside *and* inside values. An increase in $w_{jt} \in W_t$ should improve j's outside value more than her inside value, because any value from wages inside the relationship is shared with her partner. As her participation constraint may thus bind, we expect $\partial\mu_{jt}/\partial w_{jt} > 0$ and, in turn, $\partial\mu_{-jt}/\partial w_{jt} < 0$. The wage w_{jt} also affects the *partner's* inside value through sharing; a wage rise loosens the partner's constraint while a wage cut tightens it (both income effects), so $\partial\mu_{-jt}/\partial w_{jt} < 0$ like above.

Wealth a_t also enters the participation constraints, but it does not clearly favour one party unless policies or explicit agreements dictate this (e.g. prenuptial contracts).

The extent to which a participation constraint binds in response to Z_t or W_t depends on the person's inside value, which, as we established earlier, varies with the applicable Pareto weight. Suppose μ_{jt-1} is the Pareto weight at the *start* of period t, after shocks manifest but before decisions are made in the period. A relatively larger μ_{jt-1} implies a relatively larger share of marital surplus for j, thus making her outside option less desirable for a given realisation of Z_t or W_t; and vice versa. So whether j's constraint binds at the *start* of period t depends on μ_{jt-1}. Consequently, ν_{jt} and the updating of the Pareto weight upon decision making later on in the period also depend on it.

Where does μ_{jt-1} come from? The nature of decision making is such that the participation constraints are always satisfied at the updated μ_{jt} at the *end* of the period. No further updating takes place before decision making in the *following* period, therefore μ_{jt} at the *end* of t is also the applicable weight at the *start* of $t+1$. By deduction, μ_{jt-1} is therefore the weight that materialised at the *end* of $t-1$. By the time of decision making in period t, μ_{jt-1} summarises the history of the household through binding past constraints and renegotiations, which is an artifact of the recursive nature of the participation constraints. Intuitively, a given shock that improves j's outside option will not trigger a renegotiation if j has been 'happy' inside her relationship, that is, if she has historically earned a 'good' share of the marital pie. So certain shocks trigger renegotiation in certain histories but not in other. History is summarised by a single variable, μ_{jt-1}, and conditional on it, older information does not matter for decision making today (Kocherlakota, 1996).

This step-like movement of the Pareto weight in response to binding participation constraints is a well-known feature of limited commitment (Mazzocco, 2007). The couple commits to the resource allocation enacted at marriage for as long as the participation constraints remain slack. Therefore, a given allocation and Pareto weight can be quite persistent. The spouses thus fully share risk up to the point when a renegotiation takes place to satisfy a constrained spouse. After the renegotiation, the new weight may itself persist until another constraint binds. The Pareto weight thus varies cross-sectionally but also longitudinally, *within* a given family. Policies that seek to empower women, e.g. targeted cash transfers,

can thus be implemented during the relationship and may have lasting effects on the balance of power if they successfully improve women's outside options. Persistence and history are features of limited commitment that we subsequently exploit for testing.

No commitment. The Pareto weight is determined in each period given the prevailing information in the period; by construction, it is given by

$$\mu_{jt} = \mu_j(\Theta_0, W_t, Z_t, a_t), \qquad j \in \{1, 2\}, \forall t.$$

The Pareto weight varies with the initial bargaining variables Θ_0, which influence the type of game the spouses play. It also varies with new information that reveals over time, i.e. shocks to wages and distribution factors (the bargaining effects of which are typically assignable), and assets. For example, an increase in $w_{jt} \in W_t$ should empower spouse j ($\partial\mu_{jt}/\partial w_{jt} > 0$) and simultaneously weaken her partner ($\partial\mu_{-jt}/\partial w_{jt} < 0$). How precisely this is done depends on the exact bargaining game on which no assumption is made beyond cooperation.

The continuous response of the Pareto weight to contemporaneous information is a well-known feature of no commitment and cooperative bargaining (e.g. Lise and Yamada, 2019). The main implication is that the spouses cannot share risk efficiently as they cannot commit to transfer resources from one period to another. This lack of history, at least conditional on assets, makes the Pareto weight transitory in nature. It implies that policies that seek to empower women may be implemented during the relationship (e.g. by improving elements in W_t or Z_t that are assignable to women) but their effect is temporary. As soon as the policy disappears, any gains in the Pareto weight also disappear. Lack of history stems from the finite horizon of our setting, as we illustrated in section 2. This absence of history is a feature of no commitment that we exploit for testing.

State space. The policy functions derived from (6) vary with wages W_t and assets a_t, which affect the budget set in all commitment modes and, through it, the trade-off between consumption, savings, and work. The policy functions also vary with the applicable Pareto weight μ_{jt}. In all alternatives, a relatively larger μ_{jt} implies period t choices that favour j, and vice versa. Consequently, the state space is given by $\Omega_t = \{W_t, a_t, \mu_{jt}\}$, common across modes.[18] However, each commitment alternative imposes different restrictions on μ_{jt}. For example, μ_{jt} in full commitment is exclusively determined by the initial bargaining variables Θ_0. We may thus replace μ_{jt} in the state space with Θ_0, so $\Omega_t = \{W_t, a_t, \Theta_0\}$ in this case. By a similar argument, the limited commitment state space is given by $\Omega_t = \{W_t, a_t, Z_t, \mu_{jt-1}\}$

[18] In practice, only one person's weight enters the state space because of the restriction $\mu_{1t}+\mu_{2t} = constant$.

while the no commitment state space by $\Omega_t = \{W_t, a_t, \Theta_0, Z_t\}$. We show subsequently that these sets are nested, which enables us to test for the type of commitment.

3.4 Nesting and restrictions on household behaviour

To understand how the different commitment modes are related, it is useful to pool together the corresponding Pareto weights, namely

$$\begin{array}{ll} \text{full commitment:} & \mu_{jt} = \mu_j(\Theta_0) \\ \text{limited commitment:} & \mu_{jt} = \mu_j(W_t, Z_t, a_t, \mu_{jt-1}) \\ \text{no commitment:} & \mu_{jt} = \mu_j(\Theta_0, W_t, Z_t, a_t), \end{array}$$

where $\mu_{jt} = \mu_{jt-1} + \nu_j(W_t, Z_t, a_t, \mu_{jt-1}) \equiv \mu_j(W_t, Z_t, a_t, \mu_{jt-1})$ in limited commitment.

The limited commitment weight depends on its past value, which summarises the history of the household from marriage until today. If observed and accounted for, μ_{jt-1} is a sufficient statistic for the past (Kocherlakota, 1996). In practice, however, μ_{jt-1} is unobserved. From its law of motion, we can substitute the past weight recursively until $t = 0$ (marriage) to get

$$\mu_{jt} = \mu_j(W_t, Z_t, a_t,\ \mu_j(W_{t-1}, Z_{t-1}, a_{t-1},\ \mu_j(W_{t-2}, Z_{t-2}, a_{t-2}, \dots,\ \mu_j(\Theta_0)))).$$

Unconditional on its past value, μ_{jt} thus depends on the entire information set since marriage, which includes all historical wages $W_{t-\tau}$, distribution factors $Z_{t-\tau}$, and assets $a_{t-\tau}$, $\tau \in \{1, \dots, t-1\}$, that affected past participation constraints and, through them, the historical dynamics of bargaining power in the household. In other words, any variable that affects the unaccounted μ_{jt-1} must also affect the Pareto weight today. Consolidating the variables that enter the limited commitment weight and reordering the modes, we obtain

$$\begin{array}{ll} \text{full commitment:} & \mu_{jt} = \mu_j(\Theta_0) \\ \text{no commitment:} & \mu_{jt} = \mu_j(\Theta_0, W_t, Z_t, a_t) \\ \text{limited commitment:} & \mu_{jt} = \mu_j(\Theta_0, W_t, Z_t, a_t, \underbrace{W_{t-1}, Z_{t-1}, a_{t-1}, \underbrace{W_{t-2}, Z_{t-2}, a_{t-2}, \dots}_{\text{enters through } \mu_{jt-2}}}_{\text{enters through } \mu_{jt-1}}), \end{array}$$

which reflects the nesting of the *sets of variables* that matter for bargaining in each case.[19]

Contemporaneous information (information in W_t or Z_t) does not matter for the Pareto weight in full commitment but it does matter in no and limited commitment. Full commit-

[19] Let $I_t = \{W_t, Z_t, a_t\}$. To be precise, we should write $\mu_{jt} = \widetilde{\mu}_j(\Theta_0, I_t, I_{t-1}, I_{t-2}, \dots)$ in limited commitment, where $\widetilde{\mu}_j$ is the reduced form of $\mu_{jt} = \mu_j(I_t, \mu_j(I_{t-1}, \mu_j(I_{t-2}, \dots,\ \mu_j(\Theta_0))))$. To avoid unnecessary notation (namely the tilde), we reinstate μ_j as the reduced form of its structural counterpart.

ment is thus nested (in terms of the variables that matter for bargaining) within both non-full commitment alternatives, a known result since Mazzocco (2007), and current information is a natural exclusion restriction that separates full from non-full commitment.

Historical information (information in $W_{t-\tau}$ or $Z_{t-\tau}$, $\tau \geq 1$) does not matter for the Pareto weight in no (or full) commitment but it does matter in limited commitment. No commitment is thus nested within limited commitment, a new result in the literature, and history is a natural exclusion restriction that separates no from limited commitment.

The problem with this approach is that the Pareto weight is unobserved. From the theory of dynamic programming, however, the optimal labour supply policies that solve (6) are functions of the state space $\Omega_t = \{W_t, a_t, \mu_{jt}\}$. Therefore, the previous exclusion restrictions on the Pareto weight immediately become exclusion restrictions on the (typically observed) individual labour supplies h_{1t} and h_{2t} in the household, given by

$$\begin{aligned}
&\text{full commitment:} && h_{jt} = h^*_{jt}(W_t, a_t, \mu_j(\Theta_0)) \\
&\text{no commitment:} && h_{jt} = h^*_{jt}(W_t, a_t, \mu_j(\Theta_0, W_t, Z_t, a_t)) \\
&\text{limited commitment:} && h_{jt} = h^*_{jt}(\underbrace{W_t, a_t}_{\text{non-bargaining state space}}, \mu_j(\underbrace{\Theta_0, W_t, Z_t, a_t, W_{t-1}, Z_{t-1}, a_{t-1}, \ldots}_{\text{bargaining state space}})),
\end{aligned}$$

for $j \in \{1, 2\}$. This holds for all choices in the household; however, the assignability of labour supply allows us to exploit the properties of the so-called *sharing rule*.

The sharing rule (ρ_{1t}, ρ_{2t}) summarises the share of total income each spouse can devote to own expenditure in a given period. If leisure is a normal good and utility not quasi-linear (as we assume throughout), an increase in ρ_{jt} increases j's leisure and decreases her supply of labour. Moreover, there is an one-to-one increasing relationship between j's Pareto weight μ_{jt} and share ρ_{jt}, therefore $\partial h_{jt}/\partial \mu_{jt} < 0$. This also implies $\partial h_{-jt}/\partial \mu_{jt} > 0$ as bargaining power is relative. In words, an improvement in j's bargaining position increases her leisure and reduces her labour supply; in parallel, her partner's bargaining position deteriorates, which reduces his leisure and increases his labour supply. This allows us to characterise how the various variables that enter the Pareto weight affect household labour supply.

Suppose we observe at least one initial bargaining variable $\theta_0 \in \Theta_0$ and at least one time-varying distribution factor $z_t \in Z_t$; suppose that theory or intuition suggest that both θ_0 and z_t empower j. Under *limited* commitment, we must jointly observe

$$\begin{aligned}
\text{(I)}: &\quad \partial h_{jt}/\partial \theta_0 < 0 \quad \text{and} \quad \partial h_{-jt}/\partial \theta_0 > 0 \\
\text{(II)}: &\quad \partial h_{jt}/\partial z_t < 0 \quad \text{and} \quad \partial h_{-jt}/\partial z_t > 0 \\
\text{(III)}: &\quad \partial h_{jt}/\partial z_{t-\tau} < 0 \quad \text{and} \quad \partial h_{-jt}/\partial z_{t-\tau} > 0 \quad \text{for } \tau \geq 1,
\end{aligned}$$

because, as θ_0 and z_t empower j, they must decrease her labour supply and increase her partner's. This must be true also for past distribution factors in (III) because, as changes in bargaining power are persistent in limited commitment, any variable that affected bargaining power in the past will have a lasting effect on behaviour in the future. Moreover, the recursive nature of the weight (and the fact that more recent shifts in distribution factors may undo previous shifts) implies that the effects of historical variables should diminish in magnitude the further back in time we go. We return to this point subsequently.

Full commitment implies that effects (II) and (III) are absent, while no commitment implies that (III) is absent. This is a consequence of the type of information that matters for the Pareto weight in each case. Effect (I) can only be observed cross-sectionally while effects (II) and (III), if present, can be observed cross-sectionally *and* longitudinally. Finally, effect (II) of current information is present in both no and limited commitment. Therefore, testing for current information alone, as in Mazzocco (2007) and Lise and Yamada (2019), does not inform whether limited *or* lack of commitment is the right framework through which household behaviour should be analyzed.

Two final remarks are in order. First, additional distribution factors increase the number of restrictions on household labour supply. Second, wages W_t are additional time-varying variables that enter the Pareto weight outside of full commitment. As is evident in the non-bargaining state variables in h^*_{jt}, of which W_t is part, wages are not conventional distribution factors because they also affect the budget set. However, *past* wages do not enter the state space outside of bargaining in limited commitment, so they satisfy the exclusion restriction of history. As they are also assignable, we should observe effect (III) in limited commitment also through past wages. Moreover, we will show that, conditional on household income, the *partner*'s *current* wage w_{-jt} does not enter the state space outside of bargaining in no/limited commitment, so it satisfies the exclusion restriction of contemporaneous information. The *partner*'s *current* wage thus serves as an additional distribution factor, inducing analogous effects to (II) in no and limited commitment. This role of wages is appealing because wages are readily available in household data while conventional distribution factors are harder to find. With these points in mind, we now turn to our test for commitment.

4 Test for commitment

Limited commitment describes an environment in which current and historical distribution factors affect behaviour. No commitment is a special case as history does not matter whereas current distribution factors do, while full commitment is a special case of no commitment in that current distribution factors do not matter. One may thus test for commitment by

testing whether current and historical values of variables that enter the Pareto weight affect household behaviour – in this case labour supply – and with what sign.

There are two main ways to estimate the optimal policy functions h^*_{jt}, which are the objects over which we implement our test. The first approach involves the full or partial specification of preferences, expectations, and bargaining. The alternative approach leaves preferences and expectations unspecified while it only specifies the reduced form dependence of the Pareto weight on its arguments. This is inspired by Blundell *et al.* (2016) who study how wage shocks transmit into labour supply and consumption in a unitary context.[20]

While the first approach enables the recovery of the specification and the assessment of counterfactuals, its main drawback is that it requires the estimation of preferences together with bargaining. Therefore, any test for commitment is ultimately a *joint* test of commitment *and* the specification used for preferences. The second approach avoids this as it does not require the specification of preferences – consequently, it is unable to recover deep parameters or evaluate counterfactuals. As our goal here is to *test* for commitment rather than recover preferences or bargaining primitives, it is natural to follow the second path. This entails the derivation of estimable labour supply equations from the model's first order conditions and the reduced form specification of the Pareto weight, both of which we describe below.

4.1 Dynamics of household labour supply

We use the general form of the household problem in (6) to derive the static optimality conditions for male and female hours. These conditions depend on the tax/benefits function $\tau(y_t)$ through the budget constraint. It is hard to make progress without restricting τ, so we follow Heathcote *et al.* (2014) and Blundell *et al.* (2016) and approximate τ as $y^D_t \equiv \tau(y_t) \approx (1-\chi_t)y_t^{1-\kappa_t}$. Tax/benefits parameters χ_t and κ_t reflect the proportionality and progressivity of the tax and benefits system. A progressive system has a strictly positive progressivity parameter κ_t while a proportional tax system has $\kappa_t = 0$. In that case, the spouses are taxed separately at the proportionality rate χ_t.

Except a few special cases of utility, the optimality conditions are *implicit* functions of hours and cannot be directly estimated in the data. We follow Blundell *et al.* (2016) and carry out a standard log-linearisation of $-u_{j[h]}$, the marginal utility of hours, around the most recent values of consumption and hours.[21] We show in appendix B that this operation yields a closed-form expression for the growth rate of male and female hours in terms of

[20] Theloudis (2017) extends this approach to a collective setting.

[21] Unlike Blundell *et al.* (2016), we do not log-linearise the intertemporal budget constraint.

changes in the Pareto weight and other variables, given by

$$\begin{aligned}\Delta \log h_{jt} = &\underbrace{\delta_{jt} h_{jt-1}^{-1} \Delta \log(1-\chi_t)}_{\text{1: tax effects}} - \underbrace{\delta_{jt}\kappa_t s_{-jt-1} h_{jt-1}^{-1} \Delta \log y_{-jt}}_{\substack{\text{2: tax disincentives} \\ \text{from partner earnings}}} + \underbrace{\delta_{jt} h_{jt-1}^{-1} \Delta \log \lambda_t}_{\substack{\text{3: wealth and} \\ \text{income effects}}} \\ &- \underbrace{\delta_{jt}\zeta_j q_{t-1} h_{jt-1}^{-1} \Delta \log q_t}_{\text{4: consumption complementarities}} + \underbrace{\delta_{jt}(1-\kappa_t s_{jt-1}) h_{jt-1}^{-1} \Delta \log w_{jt}}_{\text{5: substitution effects}} - \underbrace{\delta_{jt} h_{jt-1}^{-1} \Delta \log \mu_{jt}}_{\text{6: bargaining effects}},\end{aligned} \tag{7}$$

where $j \in \{1,2\}$, $-j$ indicates j's partner, and Δ is the first difference between $t-1$ and t.[22] The first two terms reflect the disincentives from shifts in, respectively, the proportionality of taxes and the partner's earnings due to progressive joint taxation. The third term reflects the wealth and income effects from shifts in the marginal utility of wealth λ_t (the Lagrange multiplier on the sequential budget constraint). The fourth term captures consumption-hours complementarities. The fifth term reflects the substitution effects on labour supply from shifts in own wage w_{jt}, accounting for progressive taxation. Finally, the sixth term reflects the bargaining effects on labour supply from shifts in the Pareto weight μ_{jt}.

Parameter δ_{jt} is given by one over $\alpha_j^{-1} + \kappa_t s_{jt-1} h_{jt-1}^{-1}$, where $\alpha_j > 0$ is approximately equal to j's Frisch elasticity of labour supply scaled by his/her hours of work, and $s_{jt} \geq 0$ is j's share of family earnings. It follows that $\delta_{jt} > 0$, which helps sign most of the terms above. For example, an increase in the proportionality of taxes reduces labour supply due to tax disincentives (term 1) while an improvement in j's Pareto weight reduces his/her hours reflecting the bargaining effects we described earlier (term 6). Finally, ζ_j reflects the nature of the consumption-hours complementarity and can thus be of any sign.[23]

Terms 1 through 5 appear also in the dynamic unitary model of Blundell *et al.* (2016) while term 6 is unique to the dynamic collective model. Expression (7) is common across commitment modes in the latter case, with differences in behaviour across modes arising mostly through the last term, i.e. through the way the Pareto weight changes in each mode.[24] Then specifying an expression for the (reduced form) dependence of μ on its arguments, which we do in the next section, allows us to use (7) to test for commitment.

The *partner*'s current wage does not explicitly appear in (7) even though $w_{-jt} \in W_t$ was previously part of the non-bargaining state space of the problem. This is because, aside of bargaining, w_{-jt} only induces tax disincentives and income effects that are fully accounted

[22] We assume the progressivity tax parameter does not change between proximate periods, so $\kappa_t = \kappa_{t-1}$. This assumption is innocuous (we show this in appendix B) and makes the notation more compact.

[23] We show in appendix B that $\alpha_j = u_{j[h]}/(u_{j[hh]} \exp(-\boldsymbol{\pi}_j^{h\prime} \boldsymbol{\xi}_{jt-1}))$ and $\zeta_j = (u_{j[hq]} \exp(-\boldsymbol{\pi}_j^{q\prime} \boldsymbol{\xi}_{jt-1}))/u_{j[h]}$.

[24] Couples in limited commitment make savings choices taking into account the effects of those choices on the outside options. Therefore, the marginal utility of wealth λ behaves differently in limited commitment compared to the other modes but this is a feature we do not exploit for testing.

for by the partner's earnings and the marginal utility of wealth. Therefore, conditional on y_{-jt} and λ_t, w_{-jt} does not affect spouse j's hours outside of bargaining. This is in contrast to w_{jt} which is the price of time and thus affects hours irrespective of bargaining.

Two final remarks are due here. First, the nature of the log-linearisation is such that the outcome equation is in terms of hours growth rather than of hours levels. While (7) is fully consistent with the policy function h^*_{jt}, it is nonetheless not the policy function itself which generally disciplines the levels of hours. Second, the Euler equation in our model is very complicated because wealth enters the Pareto weight outside of full commitment.[25] Our use of the static optimality condition avoids this complication at the cost of introducing a term for the marginal utility of wealth, $\Delta \log \lambda_t$, to which we return in section 5.3.

4.2 Dynamics of the Pareto weight

Our goal is to specify the reduced form dependence of the Pareto weight on its arguments, which will enable us to take (7) to the data. Consider the structural version of the most general Pareto weight, i.e. of limited commitment, given by $\mu_{jt} = \mu_j(W_t, Z_t, a_t, \mu_{jt-1})$ with $\mu_{j0} = \mu_j(\Theta_0)$, $j \in \{1, 2\}$. To simplify the discussion, let μ_{jt} be a function of one stochastic distribution factor $z_t \in Z_t$ and the past Pareto weight only, i.e. $\mu_{jt} = \mu_j(z_t, \mu_{jt-1})$, and let μ_{j0} be a function of one initial factor $\theta_0 \in \Theta_0$. Assume without loss of generality that both factors empower spouse j, i.e. $\partial \mu_{jt}/\partial z_t > 0$ and $\partial \mu_{j0}/\partial \theta_0 > 0$. We generalise the discussion to multiple distribution factors (as well as wages and assets) in appendix C.

Suppose momentarily that $\ddot{\mu}_j(z_t, \mu_{jt-1})$ is the smooth approximation of $\mu_j(z_t, \mu_{jt-1})$. If the steps in μ are sufficiently small, $\ddot{\mu}$ will be a reasonable approximation of the true dynamics of the Pareto weight. Appendix C shows that a log-linearisation of $\mu_{jt} \approx \ddot{\mu}_j(z_t, \mu_{jt-1})$ yields

$$\Delta \log \mu_{jt} \approx e_{\mu_j,z} \Delta \log z_t + e_{\mu_j,\mu_{jL}} \Delta \log \mu_{jt-1},$$

where $e_{\mu_j,z}$ is the elasticity of μ_j w.r.t. z ($e_{\mu_j,z} > 0$ due to the assignability of z) and $e_{\mu_j,\mu_{jL}}$ is its elasticity w.r.t. the past weight ($e_{\mu_j,\mu_{jL}} > 0$ reflecting persistence in the Pareto weight in limited commitment; subscript L denotes the lag). This expression relates the contemporary shifts in μ_{jt} to the contemporary shifts in the distribution factor and the most recent historical dynamics of the weight. Due to the nature of the log-linearisation, the elasticities depend on the *past levels* of the distribution factor and thus vary in the cross-section.

Exploring the recursive nature of $\Delta \log \mu_{jt}$, we can substitute the past weight backwards until we reach $t = 0$, i.e. marriage, and $\Delta \log \mu_{j0}$ on the right hand side. $\Delta \log \mu_{j0}$ describes the formation of the initial Pareto weight at marriage, i.e. the difference between the initial

[25] Mazzocco (2007) uses a simpler Euler equation by assuming wealth does not affect the outside options.

weight μ_{j0} and a generic one available to all individuals when they first meet and start dating at $t = -1$.[26] The difference is driven by the initial distribution factor θ_0 and, as different couples differ in θ_0, $\Delta \log \mu_{j0}$ varies in the cross-section as a function of it. We adapt the simple log-linear formulation $\Delta \log \mu_{j0} = e_{\mu_j,\theta}\theta_0$, where $e_{\mu_j,\theta}$ reflects the loading factor of θ_0 onto the initial weight. We expect $e_{\mu_j,\theta} > 0$ from the assignability of θ_0.

Combining these steps (exact derivation in appendix C) yields

$$\Delta \log \mu_{jt} \approx \underbrace{\sum_{\tau=0}^{t-1} (e_{\mu_j,\mu_{jL}})^{\tau} e_{\mu_j,z} \Delta \log z_{t-\tau}}_{\substack{\text{cumulative growth in Pareto weight} \\ \text{due to time-varying distribution factors}}} + \underbrace{(e_{\mu_j,\mu_{jL}})^{t} e_{\mu_j,\theta}\theta_0}_{\substack{\text{formation of initial} \\ \text{Pareto weight} \\ \text{at marriage}}}, \tag{8}$$

where t reflects the number of periods since marriage. (8) shows that the limited commitment Pareto weight at t is the accumulation of gradual shifts in the weight over time as a result of shifts in current ($\tau = 0$) and historical ($\tau = 1, \ldots, t-1$) distribution factors z_τ, as well as θ_0 that reflects the formation of bargaining power at marriage. The effect of the distribution factor z on the current weight is given by $\partial \Delta \log \mu_{jt} / \partial \Delta \log z_{t-\tau} = (e_{\mu_j,\mu_{jL}})^{\tau} e_{\mu_j,z} > 0$. In appendix C, we show that $e_{\mu_j,\mu_{jL}} \leq 1$, so historical distribution factors have a gradually smaller effect as the length of time increases, with the rate of decay determined by $e_{\mu_j,\mu_{jL}}$.

Expression (8) encapsulates the alternative commitment modes and our nesting argument. Limited commitment has $e_{\mu_j,\mu_{jL}} > 0$, $e_{\mu_j,z} > 0$, $e_{\mu_j,\theta} > 0$, with the latter two signed by the assignability of z and θ_0. No commitment has $e_{\mu_j,\mu_{jL}} = 0$, $e_{\mu_j,z} > 0$, $e_{\mu_j,\theta} > 0$; as such no commitment is nested within limited commitment in terms of the variables that enter the Pareto weight. Full commitment has $e_{\mu_j,\mu_{jL}} = 0$, $e_{\mu_j,z} = 0$, $e_{\mu_j,\theta} > 0$; as such full commitment is nested within no commitment. Finally, the unitary model has $e_{\mu_j,\mu_{jL}} = 0$, $e_{\mu_j,z} = 0$, $e_{\mu_j,\theta} = 0$; as such the unitary model is nested within the full commitment collective model. Conditional on identifying $e_{\mu_j,\mu_{jL}}$, $e_{\mu_j,z}$, $e_{\mu_j,\theta}$, these points suggest the type of hypotheses one can formulate and assess in the data.

Informed by (8), our choice of specification for the reduced form dependence of the Pareto weight on its arguments is

$$\Delta \log \mu_{jt} \approx \sum_{\tau=0}^{t-1} \eta^{z}_{j\tau} \Delta \log z_{t-\tau} + \eta^{\theta}_{jt}\theta_0. \tag{9}$$

The $\eta^{z}_{j\tau}$'s and η^{θ}_{jt} are reduced form elasticities for the response of j's Pareto weight to

[26] Let newly met individuals start dating with the same bargaining power (an innocuous normalisation); the weight μ_{j0} of pairs that marry at $t = 0$ differs from the generic value at $t = -1$, with the difference determined by the distribution factor θ_0 realised at $t = 0$.

distribution factors: $\eta^z_{j\tau}$ captures the effect of the z factor τ periods in the past; η^θ_{jt} captures the effect of the θ initial factor t periods after marriage. With additional distribution factors (and wages and assets) affecting bargaining, the number of parameters increases considerably as we show in appendix C. Moreover, in our richest specification subsequently, we let the $\eta^z_{j\tau}$'s depend on the past levels of the distribution factors, that is $\eta^z_{j\tau} = \eta^z_{j\tau}(z_{t-\tau-1})$, to mimic the dependence of the elasticities $e_{\mu_j,z}$ and $e_{\mu_j,\mu_{jL}}$ on the past levels of the factors. A given shift in a distribution factor may thus shift the Pareto weight or not (that is, in spite of the smooth formulation in (9)), depending on the factors' historical values.

4.3 Formulation of test

Our final equation for hours combines the dynamics of household labour supply in (7) with the Pareto weight in (9). After introducing additional distribution factors $z_{1t}, z_{2t} \in Z_t$ and $\theta_{10}, \theta_{20} \in \Theta_0$, reinstating wages $w_{1t}, w_{2t} \in W_t$ and assets a_t as arguments in μ_{jt}, and pooling common terms, the combined hours equation for spouse $j \in \{1,2\}$ is given by

$$
\begin{aligned}
\Delta \log h_{jt} \;=\; & \delta_{jt} h_{jt-1}^{-1} \Delta \log(1-\chi_t) \;-\; \delta_{jt}\kappa_t s_{-jt-1} h_{jt-1}^{-1} \Delta \log y_{-jt} \\
& +\; \delta_{jt} h_{jt-1}^{-1} \Delta \log \lambda_t \;-\; \delta_{jt}\zeta_j q_{t-1} h_{jt-1}^{-1} \Delta \log q_t \\
& \underbrace{+\; \delta_{jt}(1-\kappa_t s_{jt-1} - \eta^{w_j}_{j0})}_{\beta_{j[w_{jt}]}\text{: substitution and bargaining effects of own current wage}} h_{jt-1}^{-1} \Delta \log w_{jt} \;\underbrace{-\; \delta_{jt}\eta^{w_{-j}}_{j0}}_{\beta_{j[w_{-jt}]}\text{: bargaining effect of partner's current wage}} h_{jt-1}^{-1} \Delta \log w_{-jt} \\
& \underbrace{-\sum_{\tau=1}^{t-1} \delta_{jt}\eta^{w_j}_{j\tau}}_{\beta_{j[w_{jt-\tau}]}\text{: bargaining effects of own past wages}} h_{jt-1}^{-1} \Delta \log w_{jt-\tau} \;\underbrace{-\sum_{\tau=1}^{t-1} \delta_{jt}\eta^{w_{-j}}_{j\tau}}_{\beta_{j[w_{-jt-\tau}]}\text{: bargaining effects of partner's past wages}} h_{jt-1}^{-1} \Delta \log w_{-jt-\tau} \\
& \underbrace{-\sum_{\tau=0}^{t-1} \delta_{jt}\eta^{z_j}_{j\tau}}_{\beta_{j[z_{jt-\tau}]}\text{: bargaining effects of current and past } z_j} h_{jt-1}^{-1} \Delta \log z_{jt-\tau} \;\underbrace{-\sum_{\tau=0}^{t-1} \delta_{jt}\eta^{z_{-j}}_{j\tau}}_{\beta_{j[z_{-jt-\tau}]}\text{: bargaining effects of current and past } z_{-j}} h_{jt-1}^{-1} \Delta \log z_{-jt-\tau} \\
& \underbrace{-\sum_{\tau=0}^{t-1} \delta_{jt}\eta^{a}_{j\tau}}_{\beta_{j[a_{t-\tau}]}\text{: bargaining effects of current and past assets}} h_{jt-1}^{-1} \Delta \log a_{t-\tau} \;\underbrace{-\; \delta_{jt}\eta^{\theta_j}_{jt}}_{\beta_{j[\theta_{j0}]}\text{: bargaining effect of } \theta_{j0}} h_{jt-1}^{-1}\theta_{j0} \;\underbrace{-\; \delta_{jt}\eta^{\theta_{-j}}_{jt}}_{\beta_{j[\theta_{-j0}]}\text{: bargaining effect of } \theta_{-j0}} h_{jt-1}^{-1}\theta_{-j0},
\end{aligned}
\tag{10}
$$

where $-j$ indicates j's spouse. To simplify the discussion, we introduce the parameters β_j as the reduced form coefficients on wages and distribution factors that enter j's hours. We show

in square brackets indexing β_j which variable each coefficient corresponds to; e.g., $\beta_{j[w_{-jt-\tau}]}$ is the coefficient on $\Delta \log w_{-jt-\tau}$, the partner's wage τ periods in the past.

Except one's own current wage, all other current and past wages and distribution factors enter the equation exclusively through the Pareto weight. We may thus formulate testable hypotheses on their coefficients in accordance with the alternative models. The own current wage w_{jt} enters (10) irrespective of bargaining (w_{jt} is the price of one's own hours), so $\beta_{j[w_{jt}]}$ cannot be part of these hypotheses.

Distribution factors do not affect behaviour in the unitary model, which thus has

$$\mathcal{H}_0^{\text{Unit.}}: \ \beta_{j[w_{-jt}]} = \beta_{j[w_{kt-\tau}]} = \beta_{j[z_{kt}]} = \beta_{j[z_{kt-\tau}]} = \beta_{j[\theta_{k0}]} = 0 \\ \text{for } \tau \in \{1, \ldots, t-1\} \text{ and } k \in \{1, 2\}.$$

In words, the coefficients on the partner's current wage, all past wages, all current and past distribution factors, and the initial factors at marriage, are zero in the unitary model.

In the full commitment collective model, initial distribution factors affect behaviour via the time $t = 0$ Pareto weight but later realisations of distribution factors do not. While this seems to suggest that $\beta_{j[\theta_{k0}]} \neq 0$, the coefficients on the initial factors be non-zero, this is *not* true in our formulation. Contrasting the reduced form specification of the Pareto weight in (9) with its structural counterpart in (8), it is clear that $\beta_{j[\theta_{k0}]} \equiv -\delta_{jt}\eta_{jt}^{\theta_k} = -\delta_{jt}(e_{\mu_j,\mu_{jL}})^t e_{\mu_j,\theta_k}$. Even if $e_{\mu_j,\theta_k} \neq 0$ and the initial factor structurally affects the initial weight, the past does not matter for behaviour in full commitment, which has $e_{\mu_j,\mu_{jL}} = 0$ and therefore $\beta_{j[\theta_{k0}]} = 0$. In other words, any effect of the initial distribution factor θ_{k0} on the Pareto weight today is through the recursive structure and the persistence of the weight that are features of limited commitment alone. Consequently, full commitment has

$$\mathcal{H}_0^{\text{FC}}: \ \beta_{j[w_{-jt}]} = \beta_{j[w_{kt-\tau}]} = \beta_{j[z_{kt}]} = \beta_{j[z_{kt-\tau}]} = \beta_{j[\theta_{k0}]} = 0 \\ \text{for } \tau \in \{1, \ldots, t-1\} \text{ and } k \in \{1, 2\},$$

which is the same as $\mathcal{H}_0^{\text{Unit.}}$. This is because our dynamic differences framework revolves around the *growth* of the Pareto weight, $\Delta \log \mu_{jt}$, which is observationally equivalent between full commitment and the unitary model. The main implication is that if we *fail* to reject the common null, this failure cannot inform us about the true underlying structure.

Under no commitment, past distribution factors do not matter for behaviour (this includes the initial factors at marriage for similar reasons as above) while current distribution factors do. No commitment thus has

$$\mathcal{H}_0^{\text{NC}}: \ \beta_{j[w_{kt-\tau}]} = \beta_{j[z_{kt-\tau}]} = \beta_{j[\theta_{k0}]} = 0 \qquad \text{for } \tau \in \{1, \ldots, t-1\} \text{ and } k \in \{1, 2\}.$$

Moreover, the remaining β_j's must clearly be of the correct sign: $\beta_{j[w_{-jt}]} > 0$ (the partner's wage worsens j's bargaining power and increases j's hours) while $\beta_{j[z_{kt}]}$ is signed according to the assignability of z_k.[27]

Under limited commitment, finally, all current and past distribution factors matter for behaviour, so all β_j's are different from zero. Limited commitment is the most general setting so it is without an alternative hypothesis or a conventional statistical test for it. Nevertheless, limited commitment remains testable in a conceptual sense. Clearly, all bargaining-related β_j's must be of the correct sign, disciplined by the assignability of the corresponding distribution factors. For example, $\beta_{j[w_{-jt}]} > 0$ (as in no commitment), $\beta_{j[w_{jt-\tau}]} < 0$ for $\tau \geq 1$ (own past wages improve past bargaining power and, through persistence in the Pareto weight, decrease own hours today), $\beta_{j[w_{-jt-\tau}]} > 0$ (for opposite reasons), and similarly for the many $\beta_{j[z_{kt-\tau}]}$'s and $\beta_{j[\theta_{k0}]}$'s according to the assignability of z_k and θ_k. Our test is therefore about the presence of effects from current and past wages and distribution factors, as well as, conceptually, about the sign of such effects.

The test offers many over-identifying restrictions. The obvious ones concern the *partner's* hours equation. For instance, to reject no commitment, we must reject $\mathcal{H}_0^{\text{NC}}$ as illustrated above *and* reject the analogous hypothesis for the β_{-j}'s. Another set of restrictions stems from the presence of multiple distribution factors that give rise to proportionality restrictions as in Bourguignon *et al.* (2009). From the structural form of the Pareto weight in (8), it is easy to see that the ratio of partial effects of any two concurrent distribution factors is independent of when the factors are timed, e.g. $(\partial \Delta \log \mu_{jt} / \partial \Delta \log w_{1t-\tau}) / (\partial \Delta \log \mu_{jt} / \partial \Delta \log w_{2t-\tau}) = e_{\mu_j,w_1} / e_{\mu_j,w_2}$ independent of τ. This translates into proportionality restrictions on the bargaining effects on hours, that is $\beta_{j[w_{1t-\tau}]} / \beta_{j[w_{2t-\tau}]} = \eta_{j\tau}^{w_1} / \eta_{j\tau}^{w_2}$ is the same for all $\tau \geq 1$.[28]

Two final remarks are due here. First, the hypotheses do not involve assets even if assets enter the Pareto weight outside of full commitment. Unlike wages or z_k and θ_k, wealth is non-assignable so it is not clear whom it empowers. Second, although $\beta_{j[w_{jt}]}$ is not useful for testing because the own current wage affects hours irrespective of bargaining, it turns out that we can identify $\eta_{j0}^{w_j}$, the Pareto weight coefficient on w_{jt}. This parameter must be positive in no and limited commitment, which offers an additional testable restriction.

[27] The partner's wage w_{-jt} worsens j's bargaining power so $\eta_{j0}^{w_{-j}}$, the reduced form coefficient on w_{-jt} in equation (9), must be negative. Then $\beta_{j[w_{-jt}]} \equiv -\delta_{jt} \eta_{j0}^{w_{-j}} > 0$ because $\delta_{jt} > 0$ by construction.

[28] The proportionality restrictions extend beyond this illustration. For example, the ratio of partial effects of a given distribution factor ι periods apart is independent of when the effects are exactly timed.

5 Empirical implementation

5.1 Data requirements

Our test requires panel data over at least three periods. Two periods are needed to form the concurrent *growth* of hours and distribution factors, thus test whether contemporaneous factors affect behaviour; this essentially tests for full commitment as in Mazzocco (2007). A third period is needed to form the immediately past growth of distribution factors, thus test whether history affects behaviour; this in turn separates no and limited commitment. Additional periods strengthen the test with over-identifying restrictions on the role of history.

The test requires at least one time-varying and one initial distribution factor assignable in the couple in order to assess all parts of $\mathcal{H}_0^{\text{FC}}$ and $\mathcal{H}_0^{\text{NC}}$. Assignability enables us to assess the sign of the coefficients. The partner's wage w_{-jt} is the obvious assignable time-varying factor (conditional on wealth and the partner's earnings, w_{-jt} does not affect choices outside of bargaining), so additional $z_{jt} \in Z_t$ are not strictly needed.

Given the requirement to have data on the household over a minimum of three periods, it follows that the couple must stay intact, i.e. not divorce, over this course of time. A couple may divorce later (see appendix A) but information from those periods is not part of our test since (10) describes behaviour for as long as the couple remains married.

5.2 Sample selection

We use public data from the Panel Study of Income Dynamics (Institute for Social Research, 2021). The PSID started in 1968 as an income and employment survey of a representative sample of US households and their split-offs. It was later redesigned to enable the collection also of expenditure and wealth data. As our estimating equation includes consumption and wealth controls, we focus on the period between 1999-2019 when this data is available.[29]

We focus on the Survey Research Center sample and we select married (or permanently cohabiting) couples between 21 and 65 years old, observed for at least three consecutive periods. We require complete data on earnings, hours, consumption, wealth, and demographics. Given our outcome variable, we restrict the sample to couples in which both partners participate in the labour market.[30] We return to this last selection in section 6.

Appendix table E1 summarises our baseline sample of 13,955 observations that meet these criteria. The sample conforms to expectations about income, hours, and demographics

[29]The PSID is biennial over 1999-2019; we let $\Delta x_t = x_t - x_{t-1}$ denote the first difference in x between 'current' and 'previous' periods, even though the 'previous' period refers to two calendar years in the past. We express all monetary amounts in 2018 dollars, using an 'all goods' CPI from OECD (2024).

[30]Among couples who meet all other selection criteria, 91.6% of men and 80.1% of women work for pay.

among couples in the PSID (e.g. Blundell *et al.*, 2016). For example, women work on average for 1,758 hours/year and earn \$47,843, which is about 79% of men's average hours (2,231 hours/year) but only 59% of men's earnings (\$81,742) respectively.

5.3 Estimation details and econometric issues

Modeling choices. We must address three final issues before we run the test. The first concerns the marginal utility of wealth. $\Delta \log \lambda_t$ reflects the change from $t-1$ to t in the couple's marginal utility over total income and wealth. The couple adjusts wealth endogenously in expectation of future states of the world; this reflects precautionary savings and, in limited commitment, investment in the outside options. As $\Delta \log \lambda_t$ is unobserved, we replace it with a function of the growth in household income $\Delta \log y_t$ and wealth $\Delta \log a_t$, namely $\Delta \log \lambda_t = \ell_{\Delta y} \Delta \log y_t + \ell_{\Delta a} \Delta \log a_t + \ell_y \log y_{t-1} + \ell_a \log a_{t-1}$. Since $\Delta \log \lambda_t$ may depend on the initial values of income and wealth, we also include terms for those.

The second issue concerns the hourly wage. Wages have a lifecycle component that agents typically anticipate and which is unlikely to induce bargaining between spouses. We specify the growth rate of wages as the sum of a deterministic component, anticipated at $t = 0$, and a stochastic component (see Meghir and Pistaferri, 2011, for a review of income processes). We write $\Delta \log w_{jt} = \boldsymbol{\pi}_j^{w\prime} \mathbf{x}_{jt}^w + \omega_{jt}$, where $\mathbf{x}_{jt}^w$ is a vector of demographics (e.g. age) that enter the deterministic part and ω_{jt} is the wage shock such that $\mathbb{E}_t(\omega_{jt} | \mathbf{x}_{jt}^w) = 0$. We assume that the shock is the only part that induces bargaining between the spouses, that is, the only component of wages that shifts the Pareto weight under no/limited commitment.[31]

The third issue concerns our choice of distribution factors in Θ_0 and Z_t. For Θ_0, we seek assignable factors set at $t = 0$ that remain constant thereafter; but we rarely observe the time of marriage in the PSID, which limits our options. We use two age-gap-at-marriage variables, namely $\Theta_0 = \{\mathbb{1}[age_1 << age_2], \mathbb{1}[age_1 >> age_2]\}$. The first dummy indicates the husband is younger than the wife while the second indicates he is much older.[32] The underlying premise is that youth empowers oneself as the youngest person's marriage market is typically *more* active, perhaps due to fertility concerns (Low, 2024). For Z_t, we seek assignable factors that vary over time. We follow Chiappori *et al.* (2012) and Dupuy and Galichon (2014) and use anthropometric measures, specifically the spouses' body mass index $Z_t = \{BMI_{1t}, BMI_{2t}\}$. Both choices are subject to limitations, which we discuss subsequently. Recall, however, that $z_{jt} \in Z_t$ are not strictly needed for our test (wages *are* time-varying distribution factors) so we present results without Z_t in the text and relegate the results *with* Z_t to appendix E.

[31] Nevertheless, the deterministic profile of own wages affects the gradient of hours through the own wage term 5 in equation (7). We show this point in detail in appendix D.

[32] We consider the husband older if he is at least 4 years older than the wife.

Specifications and heterogeneity. After implementing these modeling choices within (10), the final equation for hours of spouse $j \in \{1,2\}$, abstracting from $Z_t = \{BMI_{1t}, BMI_{2t}\}$, is given in compact form by

$$\begin{aligned}\Delta \log h_{jt} = \Big\{ & b_{j[0]} + \mathbf{b}'_{j[x^w_{jt}]}\mathbf{x}^w_{jt} \\ & + \beta_{j[w_{jt}]}\omega_{jt} + \beta_{j[w_{-jt}]}\omega_{-jt} + \sum_{\tau=1}^{t-1}\beta_{j[w_{jt-\tau}]}\omega_{jt-\tau} + \sum_{\tau=1}^{t-1}\beta_{j[w_{-jt-\tau}]}\omega_{-jt-\tau} \\ & + \beta_{j[young_j]}\mathbb{1}[age_j << age_{-j}] + \beta_{j[young_{-j}]}\mathbb{1}[age_j >> age_{-j}] \\ & + b_{j[\Delta y_t]}\Delta \log y_t + b_{j[\Delta a_t]}\Delta \log a_t + b_{j[y_{t-1}]}\log y_{t-1} + b_{j[a_{t-1}]}\log a_{t-1} \\ & + b_{j[\Delta y_{-jt}]}s_{-jt-1}\Delta \log y_{-jt} + b_{j[\Delta q_t]}q_{t-1}\Delta \log q_t + \sum_{\tau=1}^{t-1}\beta_{j[\Delta a_{t-\tau}]}\Delta \log a_{t-\tau}\Big\} \times h^{-1}_{jt-1}.\end{aligned} \tag{11}$$

Appendix D constructs this equation step by step.

We estimate three gradually richer specifications. The *first* is the reduced form linear regression of $\Delta \log h_{jt}$ on the right hand side variables in (11), treating the coefficients as constant in the cross-section/over time. Estimation via OLS delivers estimates of the reduced form bargaining effects of wages and distribution factors, enabling quick testing of the commitment hypotheses. This simplest form of our test thus affirms that the test is easy to implement in reduced form without imposing or estimating preferences.

The pure reduced form neglects the dependence of the coefficients on hours and earnings through $\delta_{jt} = 1/(\alpha_j^{-1} + \kappa_t s_{jt-1} h^{-1}_{jt-1})$. Our *second* specification estimates (11) respecting the underlying structure of the coefficients. We fix the tax progressivity parameter $\kappa_t = 0.185$ (Blundell *et al.*, 2016) and identify δ_{jt} through $b_{j[y_{-jt}]}$; see appendix D for details. This enables us to estimate α_j (a scaled Frisch labour supply elasticity) and all bargaining parameters η_j, including $\eta^{w_j}_{j0}$ that describes how the Pareto weight shifts with own current wage w_{jt}.

The elasticities of the Pareto weight depend on the immediate past levels of the distribution factors (section 4.2). We reflect this in our *third* specification by letting the η_j's depend on such past levels. There is no reason to believe that all couples feature the same degree of commitment: in reality, some couples may commit fully while others may not. In principle, we could test for commitment on a household-by-household basis, allowing each household their own degree of commitment, i.e. a household-specific η_j. Albeit theoretically appealing, this requires long time series that we clearly lack. To get around this, we pool households together and estimate a form of aggregate bargaining effect across households, one that depends, however, on the past values of wages and distribution factors. A distribution factor

may thus induce *heterogeneous* bargaining effects for given hours and earnings, depending on the distribution factors' historical values.

We estimate all three specifications initially in the baseline sample of households that we observe for at least three consecutive periods. Subsequently, we estimate them again over a smaller sample of households observed for at least four consecutive periods. This strengthens our test with additional restrictions on the role of history. We do not go beyond four periods because we run into small samples.

Empirical strategy. Unlike the first specification, estimation in the second and third cases requires GMM because of the non-linear structure of δ_{jt}. GMM estimation of the full structure of (11) is slow due to the dimensionality of the wage covariates $\mathbf{x}_{jt}^{w}$ and the household observables that we empirically account for (appendix A shows the role of demographics in the model explicitly). To avoid this, we run a first stage regression to net $\Delta \log h_{jt}$ of the demographics. We then estimate the remaining terms in a second stage using residual hours on the left hand side. This two-step estimation is similar to Blundell *et al.* (2016); a single step delivers similar point estimates, albeit much more slowly.

Our empirical strategy proceeds as follows. First, we regress wage growth $\Delta \log w_{jt}$ on observables to obtain the wage shock ω_{jt}. Second, we regress hours growth $\Delta \log h_{jt}$ on inverse past hours and taste/wage observables (times inverse past hours) to obtain residual hours consistent with the first line in (11).[33] Third, we regress residual hours on all other variables in (11), which allows us to estimate the various bargaining effects. Whenever there are over-identifying restrictions, we use multiple moments and a diagonal weighting matrix. We estimate heteroskedasticity-consistent standard errors, with the household serving as a cluster. We discuss measurement error in wages and hours below.

6 Results

Reduced form specification. The first results concern the bargaining effects of wages from the reduced form (first) specification of the commitment test. This is the simplest form of our test, one that can quickly inform about commitment in a sample of observations.

Columns 1 & 2 of table 1 show the wage terms in the male and female hours equations in the baseline sample of households observed for at least three periods.[34] Three points

[33]The first stage observables for wages and hours include dummies for year, year of birth, education, race, region, number of family members (and its change over time), number of children (and its change over time), the presence of income recipients other than the main couple (present and past), and the presence of outside dependents (present and past). We also include education-year, race-year, and region-year interactions.

[34]The number of observations is smaller than 13,955 as the estimating equations are in first difference.

emerge. First, the shock to the partner's wage at t enters significantly positively in the male equation, and it is small, negative, and insignificant in the female equation ($\beta_{j[w_{-jt}]}$).[35] This term should be zero in full commitment but positive in the other regimes. Under no and limited commitment, an increase in the partner's wage empowers the partner, thus worsens one's own relative bargaining position and increases one's own labour supply.[36] The significant positive effect on the male side is thus indicative of non-full commitment.

Second, the shocks to wages at $t-1$, both to own and partner wages, enter both equations mostly significantly and with the sign predicted by limited commitment. Under full or no commitment past events do not matter for current behaviour so $\beta_{j[w_{jt-1}]}$ and $\beta_{j[w_{-jt-1}]}$ should be zero. By contrast, history matters in limited commitment and past events that shift past bargaining power have lasting effects on behaviour in a specific way. This is precisely what we see here: own past shocks reduce own labour supply while the partner's shocks increase it. This is consistent with power shifts under limited commitment, in which favourable shocks improve the bargaining power of the spouse that receives them and, through persistence in the Pareto weight, reduce own future labour supply and increase the partner's.[37]

Third, the initial distribution factors enter both equations significantly and mostly with the sign limited commitment postulates. The dummy for being the younger spouse consistently enters negatively ($\beta_{j[young_j]}$) while the dummy for being the older spouse enters positively among women, the expected sign under limited commitment, but negatively among men ($\beta_{j[young_{-j}]}$). We introduced these age-gap-at-marriage dummies to reflect the idea that youth empowers oneself through a relatively more active marriage market. Yet, it is unclear if these variables are distribution factors in the true sense or if they also affect tastes. But it is hard to explain from a tastes argument alone why being the youngest reduces hours (the young typically work more), thus explain why $\beta_{j[young_j]} < 0$ as we find here.

Columns 3 & 4 show the wage terms in the smaller sample of households observed for at least four consecutive periods. The extra period allows us to introduce the wage shocks at $t-2$ and test additional restrictions on the role of history. While significance in a statistical sense is reduced (we are estimating a more flexible model over a smaller sample), the effects of wage shocks at t and $t-1$ and the effects of the initial factors remain qualitatively unchanged from the baseline. In addition, the shocks at $t-2$ enter both equations consistent with limited commitment. The husband's older shocks reduce his hours and increase the wife's

[35] The shock to *own* wage at t enters negatively in both equations ($\beta_{j[w_{jt}]}$). This measures an aggregate of substitution and bargaining effects of own wages so $\beta_{j[w_{jt}]}$ is not used as part of the commitment test.

[36] Recall that we control for family income, wealth, and partner earnings, so the partner's wage effect cannot be interpreted as an income, wealth, or joint taxation effect.

[37] Recall that we control for family income and wealth (i.e. realisation of the state space), so we account for income and wealth effects that past wage shocks induce on future behaviour. Such effects would operate with the same sign across male and female hours, while the coefficient $\beta_{j[w_{kt-1}]}$ clearly flips its sign over j.

Table 1: *Commitment Test – Summary of Reduced Form Results*

				≥ 3 periods (current & past shocks)		≥ 4 periods (current, past, & older shocks)	
Dependent variable:				(1) Male	(2) Female	(3) Male	(4) Female
$\Delta \log h_{jt}$	FC	NC	LC	$j=1$	$j=2$	$j=1$	$j=2$
Current shocks (t)							
$\beta_{j[w_{jt}]}$	.	.	.	−30.674	−10.276	−34.751	−15.996
				(10.657)	(3.516)	(19.945)	(12.846)
$\beta_{j[w_{-jt}]}$	0	+	+	52.842	−2.564	39.383	−11.350
				(26.149)	(6.244)	(30.972)	(18.735)
Past shocks $(t-1)$							
$\beta_{j[w_{jt-1}]}$	0	0	−	−13.856	−7.338	−11.018	−5.031
				(5.284)	(3.441)	(10.115)	(10.305)
$\beta_{j[w_{-jt-1}]}$	0	0	+	41.045	6.011	29.491	−6.877
				(20.351)	(7.113)	(34.599)	(17.107)
Older shocks $(t-2)$							
$\beta_{j[w_{jt-2}]}$	0	0	−			−32.739	−9.643
						(16.697)	(6.754)
$\beta_{j[w_{-jt-2}]}$	0	0	+			2.417	24.670
						(28.806)	(15.157)
Initial distribution factors $(t=0)$							
$\beta_{j[young_j]}$	0	0	−	−28.064	−14.680	−96.972	−47.766
				(26.590)	(6.370)	(33.136)	(11.893)
$\beta_{j[young_{-j}]}$	0	0	+	−80.557	10.376	−128.468	20.788
				(30.226)	(4.582)	(40.310)	(10.499)
p value for $\mathcal{H}_0^{\text{FC}}$				0.000	0.000	0.001	0.000
p value for $\mathcal{H}_0^{\text{NC}}$				0.001	0.000	0.001	0.000
Observations				8,513		6,028	

Notes: The table reports the coefficients on wages from the reduced form (first) specification of the commitment test. The full results including the terms of the test that we do not explicitly show here appear in online appendix table E2. Standard errors clustered at the household level are in brackets.

while, symmetrically, the wife's older shocks reduce her hours and raise the husband's. These effects, which are consistent with power shifts under limited commitment as in the case of the $t-1$ shocks above, should be absent in either full or no commitment.[38]

Regardless of the length of history, we reject full commitment (therefore also the unitary model) and no commitment at least at the 0.1% significance level in all cases.[39]

[38]Wage shocks may be correlated over time or across spouses. This does not jeopardise our test as we explicitly include the male & female shocks over multiple periods as standalone regressors (appendix G).

[39]In principle, we should test the null hypotheses *jointly* across male and female equations. For simplicity,

The large coefficients of table 1 are the result of h_{jt-1}^{-1} multiplying the right hand side of (11). We can thus not interpret the coefficients as the elasticities of hours with respect to wage shocks. It is straightforward to calculate the partial effects of wages but, in the interest of brevity, we only report this in appendix E for our richest specification below.

Structural specification. The second set of results accounts for the underlying structure of the coefficients in (11), which the reduced form results neglect. This allows us to estimate α_j, a scaled Frisch elasticity of labour supply, and $\eta_{j0}^{w_j}$, the Pareto weight coefficient on *own current* wage. It also accounts for the dependence of the coefficients on earnings and hours, thus relaxing the restriction of homogeneity of effects across households.[40]

Results across columns 1 & 2 (three periods) and 3 & 4 (four periods) of table 2 paint a similar picture. We report four main points. First, the own current wage shock improves one's Pareto weight ($\eta_{j0}^{w_j} > 0$) while the partner's current shock worsens it ($\eta_{j0}^{w_{-j}} < 0$). This contradicts full commitment but it is consistent with the other two modes. Second, own past shocks from $t-1$ and $t-2$ empower oneself ($\eta_{j1}^{w_j} > 0$, $\eta_{j2}^{w_j} > 0$), while the partner's past shocks either weaken oneself or leave him/her untouched ($\eta_{j1}^{w_{-j}} \leq 0$, $\eta_{j2}^{w_{-j}} \leq 0$). Under full or no commitment, past shocks should not affect current bargaining power. Statistical significance is generally low but any parameter that *is* significant has the sign that limited commitment requires. Third, being the younger spouse empowers oneself ($\eta_{jt}^{young_j} > 0$) while being the older spouse produces mixed results as in the reduced form previously. Fourth, $\alpha_j/\mathbb{E}(h_{jt-1})$ is approximately equal to the Frisch elasticity of labour supply evaluated at average hours; we estimate it at 0.55-1.02 for men, 0.33-0.72 for women, generally in line with the literature.[41] In all cases, we reject full and no commitment at conventional significance levels.[42]

The disproportionately large Pareto weight elasticity with respect to own vs the partner's *current* wage ($\eta_{j0}^{w_j}$ vs $\eta_{j0}^{w_{-j}}$), in contrast to the similar magnitudes of the analogous elasticities with respect to *past* wages, questions the proportionality properties of distribution factors. To assess this, we impose proportionality in the Pareto weight elasticities, which also enables us to estimate the elasticity of μ_j with respect to its past value.[43] The results in appendix table E4 are in line with the earlier results without providing a worse overall fit. We reject

we estimate the equations separately (no cross-equation restrictions) so we also conduct testing separately.

[40]The coefficients in (11) are functions of $\delta_{jt} = 1/(\alpha_j^{-1} + \kappa_t s_{jt-1} h_{jt-1}^{-1})$, so they depend on the underlying past earnings shares s_{jt-1} and hours h_{jt-1}. Appendix D provides clarity on this point.

[41]α_j/h_{jt-1} is *exactly* equal to the Frisch labour supply elasticity if preferences are separable between hours and consumption. Otherwise, the Frisch elasticity is a function of α_j/h_{jt-1} and the extent of non-separability (Blundell *et al.*, 2016). The larger male elasticity, while not uncommon, e.g. Wu and Krueger (2021), may partly reflect a different degree of consumption-leisure complementarity between men and women.

[42]It is straightforward to recast the commitment hypotheses in terms of the Pareto weight elasticities η_j.

[43]We re-estimate the model subject to $\eta_{j\tau}^{w_k} = (e_{\mu_j,\mu_{jL}})^\tau e_{\mu_j,w_k}$, for $j,k \in \{1,2\}$ and $\tau \in \{0,1,2\}$, as per the log-linearised Pareto weight in (8).

Table 2: *Commitment Test – Summary of Structural Results*

				≥ 3 periods (current & past shocks)		≥ 4 periods (current, past, & older shocks)	
Dependent variable:				(1) Male	(2) Female	(3) Male	(4) Female
$\Delta \log h_{jt}$	FC	NC	LC	$j=1$	$j=2$	$j=1$	$j=2$
Pareto weight elasticities w.r.t current shocks ($\tau=0$)							
$\eta_{j0}^{w_j}$	0	+	+	1.020	1.007	1.004	1.006
				(0.010)	(0.010)	(0.011)	(0.012)
$\eta_{j0}^{w_{-j}}$	0	−	−	−0.034	−0.003	−0.026	−0.006
				(0.022)	(0.012)	(0.017)	(0.018)
Pareto weight elasticities w.r.t shocks 1 period in the past ($\tau=1$)							
$\eta_{j1}^{w_j}$	0	0	+	0.012	0.009	0.010	0.002
				(0.006)	(0.008)	(0.006)	(0.010)
$\eta_{j1}^{w_{-j}}$	0	0	−	−0.032	0.000	−0.031	0.002
				(0.022)	(0.019)	(0.020)	(0.015)
Pareto weight elasticities w.r.t shocks 2 periods in the past ($\tau=2$)							
$\eta_{j2}^{w_j}$	0	0	+			0.010	0.009
						(0.012)	(0.005)
$\eta_{j2}^{w_{-j}}$	0	0	−			−0.023	−0.026
						(0.018)	(0.016)
Pareto weight elasticities w.r.t initial distribution factors							
$\eta_{jt}^{young_j}$	0	0	+	0.046	0.025	0.062	0.038
				(0.024)	(0.011)	(0.024)	(0.014)
$\eta_{jt}^{young_{-j}}$	0	0	−	0.096	−0.023	0.077	−0.021
				(0.035)	(0.010)	(0.037)	(0.009)
Frisch elasticity $\alpha_j/\mathbb{E}(h_{jt-1})$				0.545	0.330	1.017	0.719
				(0.164)	(0.136)	(0.347)	(0.263)
p value for $\mathcal{H}_0^{\text{FC}}$				0.000	0.000	0.000	0.000
p value for $\mathcal{H}_0^{\text{NC}}$				0.025	0.032	0.052	0.049
Observations				8,513		6,028	

Notes: The table reports the Pareto weight elasticities with respect to wages from the structural (second) specification of the commitment test. The full results including the terms that we do not explicitly show here appear in online appendix table E3. Standard errors clustered at the household level are in brackets.

full and no commitment. We estimate $e_{\mu_j,\mu_{jL}}$ at about 0.015 (statistically significant at this value), revealing a positive but weak association between current and past Pareto weight.

Heterogeneity. The third set of results allows for heterogeneity in the Pareto weight elasticities η_j through their dependence on the *past levels* of wages. Along with the dependence of the coefficients on earnings and hours (structural specification), the third specifi-

cation introduces vast heterogeneity in the bargaining effects of wages. As households differ in their wages, earnings, and hours, we calculate household-specific partial effects of wage shocks, namely $\partial \Delta \log h_{jt} / \partial \omega_{kt-\tau}$ for $j, k \in \{1, 2\}$ and $\tau \in \{0, 1, 2\}$. Figure 1 plots the partial effects for all households in the sample; the left (right) graphs show the effects of own shocks (partner's shocks); each row corresponds to shocks from a different period. We report key moments and parameter estimates in appendix tables E5 and E6.

The average partial effects of older shocks (shocks from period $t-2$ in figure 1c) are $\mathbb{E}(\partial \Delta \log h_{jt} / \partial \omega_{jt-2}) < 0$ and $\mathbb{E}(\partial \Delta \log h_{jt} / \partial \omega_{-jt-2}) > 0$. Away from the average, most households exhibit negative or zero labour supply effects from own shocks (favourable *own* shocks empower oneself) and positive or zero effects from partner shocks (favourable *partner* shocks weaken oneself). These effects are consistent with power shifts in limited commitment, in which good past shocks improved the bargaining power of the spouse that received them and, through persistence in the Pareto weight, shift future labour supply. It is remarkable that the effects survive 4 calendar years (the time that lapses from $t-2$ to t in the PSID), the controls for all subsequent wage shocks, and the multiple wealth and income controls.

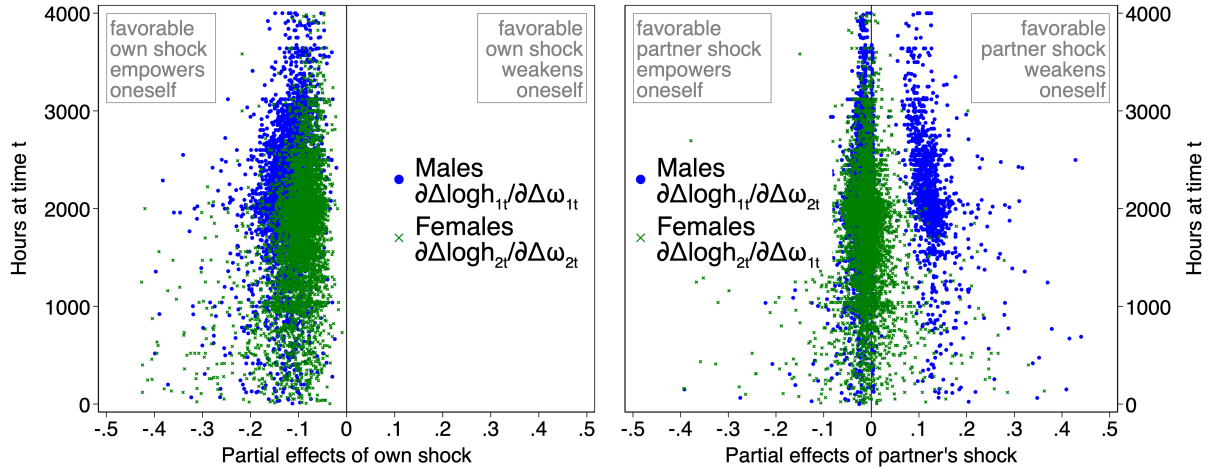
We estimate the average partial effects of immediately past shocks (from $t-1$ in figure 1b) at $\mathbb{E}(\partial \Delta \log h_{2t} / \partial \omega_{2t-1}) < 0$ and $\mathbb{E}(\partial \Delta \log h_{1t} / \partial \omega_{2t-1}) > 0$, exactly as limited commitment requires. The other two average partial effects seem to contradict limited commitment if taken at face value. $\mathbb{E}(\partial \Delta \log h_{1t} / \partial \omega_{1t-1})$ is weakly positive but the underlying parameters are not statistically significant; $\mathbb{E}(\partial \Delta \log h_{2t} / \partial \omega_{1t-1})$ is weakly negative but the majority of households exhibit *positive* effects exactly as limited commitment postulates.

The partial effects of the partner's current shocks (figure 1a) are mostly positive among men, consistent with non-full commitment, but spread around zero among women.

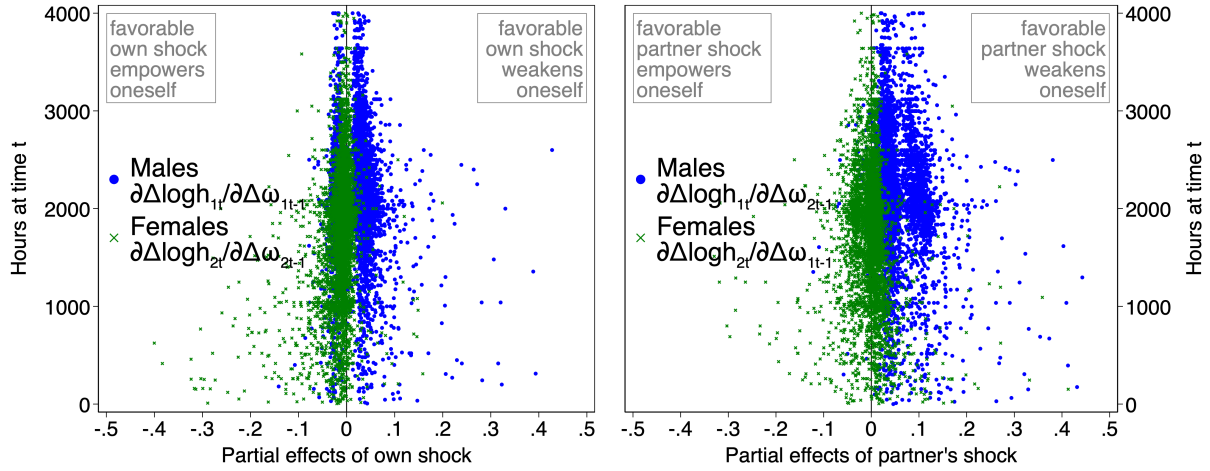
The partial effects are in fact elasticities of labour supply w.r.t wage shocks. As the effects of past shocks lie on average between 0.01-0.05 in absolute value, a 10% wage change shifts hours by 0.1%-0.5%. Past shocks exclusively induce *bargaining* effects, so this gives a sense of the average magnitude of bargaining. However, there is large heterogeneity; clearly, many couples exhibit full commitment (null bargaining effects), while others exhibit limited commitment. Appendix table E7 relates this heterogeneity to education homophily, length of marriage, and the gender of the main breadwinner in the couple.

Additional distribution factors. We use $Z_{t-\tau} = \{BMI_{1t-\tau}, BMI_{2t-\tau}\}$, $\tau \in \{0, 1, 2\}$, as a means of over-identification. We report results in appendix tables E2 (reduced form), E3 (structural), and E4 (proportionality restrictions), along with a discussion of the limitations of BMI in this context. The bargaining effects of wages are similar to the baseline; the bargaining effects of BMI are also in line with limited commitment.

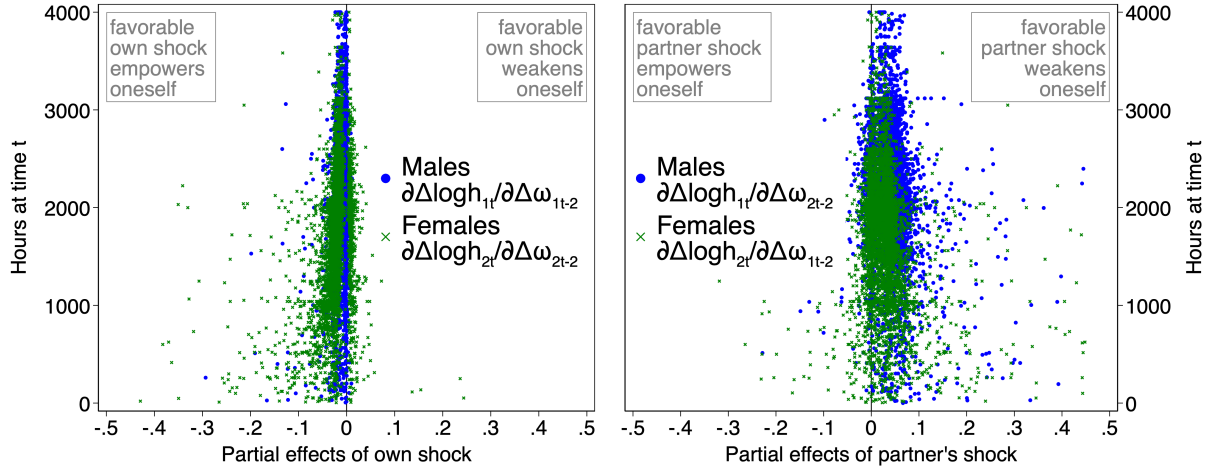
Figure 1: *Partial Effects of Wage Shocks*



(a) $\partial \Delta \log h_{jt}/\partial \omega_{kt}$, partial effects of current shocks



(b) $\partial \Delta \log h_{jt}/\partial \omega_{kt-1}$, partial effects of immediately past shocks



(c) $\partial \Delta \log h_{jt}/\partial \omega_{kt-2}$, partial effects of older shocks

Notes: The figure plots the partial derivative of hours w.r.t wage shocks, $\partial \Delta \log h_{jt}/\partial \omega_{kt-\tau}$, $j, k \in \{1, 2\}$, $\tau \in \{0, 1, 2\}$, across couples observed for ≥ 4 periods. The parameter estimates are in appendix table E6.

Measurement error. Measurement error in hours & wages biases our estimates towards zero (full commitment), but it generally preserves the true sign of the coefficients on the *partner*'s shocks. We consistently reject full and no commitment, even if we focus only on the bargaining effects from the *partner*'s shocks. Appendix F provides details.

Extensions. For simplicity, our model abstracted from several features at the intersection of household decision making and commitment that may matter for the formulation of our test. Appendix G presents extensions to home production, private and joint leisure, private consumption, human capital, labour market participation, nonlinear wage effects, and costly renegotiation. We also discuss why we have purposefully not imposed restrictions on the stochastic process governing the wage shocks. We argue (and support with empirical results whenever possible) that none of these issues jeopardise our test for commitment.

7 Conclusions

Many policies that entail an intertemporal aspect (from conditional cash transfers, to divorce and property division rules, to child support) cannot be analyzed without reference to some model of household behaviour. A particularly important aspect is commitment, the extent to which the spouses commit to a plan that disciplines their behaviour in the future.

In this paper we ask *to what extent spouses commit*. Using a lifecycle collective model, we characterise household behaviour in three prominent alternative regimes: full, limited, and no commitment. Current and past news affect the allocation of resources differently in each case. Full commitment is nested within no commitment in terms of the information that matters for such allocation, which in turn is nested within limited commitment. Nesting and natural exclusion restrictions from contemporaneous and historical information allows us to devise a test that distinguishes between the three alternatives.

We consistently reject full and no commitment in the PSID. By contrast, we find strong evidence for limited commitment. Favourable current and past shocks from 2 $(t-1)$ and 4 $(t-2)$ years ago decrease one's labour supply and increase their partner's. We show that these asymmetric effects are consistent with a power shift in the couple, in which favourable news improve the bargaining power of the recipient spouse and, through that, reduce own labour supply and raise the partner's.

To the best of our knowledge, ours is the first paper that brings the three commitment alternatives together, contrasts their implications for behaviour, and proposes a test that distinguishes between them. The insights we provide are applicable to a broad range of risk sharing arrangements, such as risk sharing among village households.

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